



Estimation of pure component properties. Part 4: Estimation of the saturated liquid viscosity of non-electrolyte organic compounds via group contributions and group interactions

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ABSTRACT

A new group contribution method for the prediction of pure component saturated liquid viscosity has been developed. The method is an extension of the pure component property estimation techniques that we have developed for normal boiling points, critical property data, and vapour pressures. Predictions can be made from simply having knowledge of the molecular structure of the compound. In addition, the structural group definitions for the method are identical to those proposed for estimation of saturated vapour pressures. Structural groups were defined in a standardized form and fragmentation of the molecular structures was performed by an automatic procedure to eliminate any arbitrary assumptions. The new method is based on liquid viscosity data for more than 1600 components. Results of the new method are compared to several other estimation methods published in literature and are found to be significantly better. A relative mean deviation in viscosity of 15.3% was observed for 813 components (12,139 data points). By comparison, the Van Velzen method, the best literature method in our benchmarking exercise produced a relative mean deviation of 92.8% for 670 components (11,115 data points). Estimation results at the normal boiling temperature were also tested against an empirical rule for more than 4000 components. The range of the method is usually from the triple or melting point to a reduced temperature of 0.75–0.8. Larger than average deviations were observed in the case of molecules with higher rotational symmetry, but no specific correction of this effect was included in this method.

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1. Introduction

Due to the importance of reliable information on liquid viscosity data for many practical applications, numerous researchers have worked on the subject. The literature concerning liquid viscosity is therefore quite extensive. Many attempts have been made to correlate and estimate the viscosity of saturated or compressed liquids as a function of temperature, pressure, and chemical constitution. Theoretical approaches have, however, not been sufficiently successful, and at present there is no theory available that allows the estimation of liquid viscosity within the required accuracy.¹

In addition, the various theoretical approaches do not sufficiently link liquid viscosity to a set of molecular properties in a similar way, as for example, gas viscosity is linked to molecular cross-section, which itself can be expressed as function of collision energy (temperature). These theoretical approaches are therefore out of the scope of this work and will not be discussed any further. A brief review of correlation methods, as well as empirical estimation approaches will be presented below. The improved approach for the estimation of liquid viscosity presented in this paper is based on our previous work on normal boiling temperatures [1,2], critical property data [3] and vapour pressures [4,5]. As in the previous work, the Dortmund Data Bank (DDB [6]) was employed as the main source for experimental data.

Even though the exact mechanisms governing liquid viscosity and vapour pressure are dissimilar, there are several similarities between these properties for a component:

- The energy required to remove a component from the liquid phase into the vapour phase or to break an existing structure of the liquid (in order to move liquid layers in opposite directions or with

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¹ In the well-known "Properties of Gases and Liquids", Poling, Prausnitz, and O'Connell point out that "little theory has been shown to be applicable to estimating liquid viscosities".

different velocity) is to a great part dependent on intermolecular attraction.

- As observed by many researchers and reviewed and extended by Smith et al. [7], the viscosity at the normal boiling temperature usually falls into a rather narrow range suggesting a link between these two properties.
- The energy required for evaporation or displacement of liquid layers is supplied by the available thermal energy RT . Thus, both vapour pressure and viscosity approximately obey an equation of the form $f(T) = \exp(A - (B/T))$.

As temperature increases, the vapour pressure increases, while viscosity decreases. Thus volatility (vapour pressure) would better compare to fluidity (the reciprocal of viscosity).

Major dissimilarities affecting the development of estimation methods between liquid viscosity and vapour pressure lie in the availability and type of experimental information for both properties:

- For the temperature range employed in this work, there is less than a third of the amount of experimental data available for liquid viscosity as compared to vapour pressure. It was therefore an advantage to develop a vapour pressure model before starting on liquid viscosity. Consequently, knowledge obtained from the development of the vapour pressure estimation method [4] proved to be important here. It was assumed that the same molecular properties determine, in different ways, vapour pressure and viscosity. Therefore, the exact same differentiation of structural groups that was required for vapour pressure estimation was also required for viscosity estimation.
- A large amount of vapour pressure data is available at a reference pressure of 1 atm (the normal boiling temperature) providing a convenient reference point. Viscosity data are often available at 25 °C. After several unsuccessful developments within this work it had to be concluded that a varying viscosity value at a fixed temperature is not a useful reference.

Vapour pressure data are needed for a variety of chemical engineering and thermodynamic calculations. These data are the main factor determining the distribution of a component between the liquid and vapour phase and therefore the key property for the design of distillation columns. Liquid viscosity data on the other hand are needed for the design of fluid transport and mixing processes (pipes, pumps, stirred reactors, etc.) and have a direct and large effect on heat transfer coefficients (heat exchangers, conduction processes, etc.) and diffusion coefficients (macro-kinetic in chemical reactors). The accuracy required of the calculated viscosity, however, is far less than that required of vapour pressure. Both the amount and quality of liquid viscosity data in literature is lower than for the case of vapour pressures. Current available estimation methods for liquid viscosity are generally of poor quality.

2. General behavior and available methods

If a shearing stress τ is applied to a unit area of a confined fluid, the fluid will move with a velocity gradient $\partial u / \partial y$ such that its maximum velocity is at the point where the stress is applied. Now, if the local shear stress per unit area at any point is divided by the velocity gradient, the ratio obtained is defined as the viscosity of the fluid. Fluids, for which the shearing stress depends linearly on the velocity gradient, are called Newtonian fluids. These fluids obey the equation:

$$\tau = \eta \frac{\partial u}{\partial y} \quad (1)$$

Only Newtonian fluids will be considered in this work.

The viscosity of gases at low densities and sufficiently high temperatures can often be described by a simple equation taking into account the mean free path and transported momentum difference (Boltzmann equation). The liquid viscosity on the other hand is governed by a different mechanism, and thus, is out of the scope of the Boltzmann equation. Besides being significantly larger, liquid viscosity shows temperature dependence opposite to that of gases. In addition, it shows significant density dependence which is not present in gases. Models for the interpretation of liquid viscosity range from simplified models such as Eyring's activated state theory and its successive modifications to approaches like Enskog's hard sphere theory, and finally include rigorous mechanical approaches in the form of the distribution function or time-correlation function methods. These types of methods mostly produce unsatisfactory results and will not be discussed further in this work.

For the correlation of liquid viscosity, similar equations can be used as in the case of vapour pressure. As viscosity diverges near the critical point, correlations employing the critical point as reference must use a hypothetical critical viscosity value.

The most simple correlation equation was first proposed by de Guzman [8], but is more commonly known as the Andrade equation:

$$\ln \frac{\eta}{\eta_{\text{ref}}} = A + \frac{B}{T} \quad (2)$$

Vogel [9] proposed another variation by the introduction of a third constant similar to the Antoine equation for vapour pressures:

$$\ln \frac{\eta}{\eta_{\text{ref}}} = A + \frac{B}{T + C} \quad (3)$$

Porter [10] was the first to draw attention to the relationship between liquid viscosities and vapour pressures, when he showed that the logarithm of viscosity for mercury and water depends more linearly on the logarithm of vapour pressure than on the inverse temperature. This provides a good argument for the assumption that liquid viscosity and vapour pressure are influenced in a similar way. Drucker [11] proposed an analytic formulation of this relation:

$$\ln \eta = A + B \ln P \quad (4)$$

However, Drucker reported that large deviations from Eq. (4) were observed for strongly associating liquids. During this work we found that the constants in the Drucker equation did not follow group contribution as well as the Vogel parameters (with the parameter C set to a certain value).

As in the case of vapour pressure, several more flexible equations are available for data correlation, but their parameters are usually more difficult to estimate due to stronger intercorrelation.

A number of group contribution methods for the estimation of saturated liquid viscosity are available in literature. The methods considered for comparison in this work are given in Table 1. A detailed description of these methods together with equations and group parameter tables is given by Nannoolal [17]. Further methods like those of Bhethanabotla [18] and Przedziecki and Sridhar [19] and methods with a limited range of applicability have not been evaluated.

Table 1

Group contribution methods for the estimation of saturated liquid viscosity considered in this work.

Year	Reference
1972	Van Velzen et al. [12]
1974	Orrick and Erbar [13]
1985	Skubla [14]
1987	Joback and Reid [15]
1992	Sastri and Rao [16]

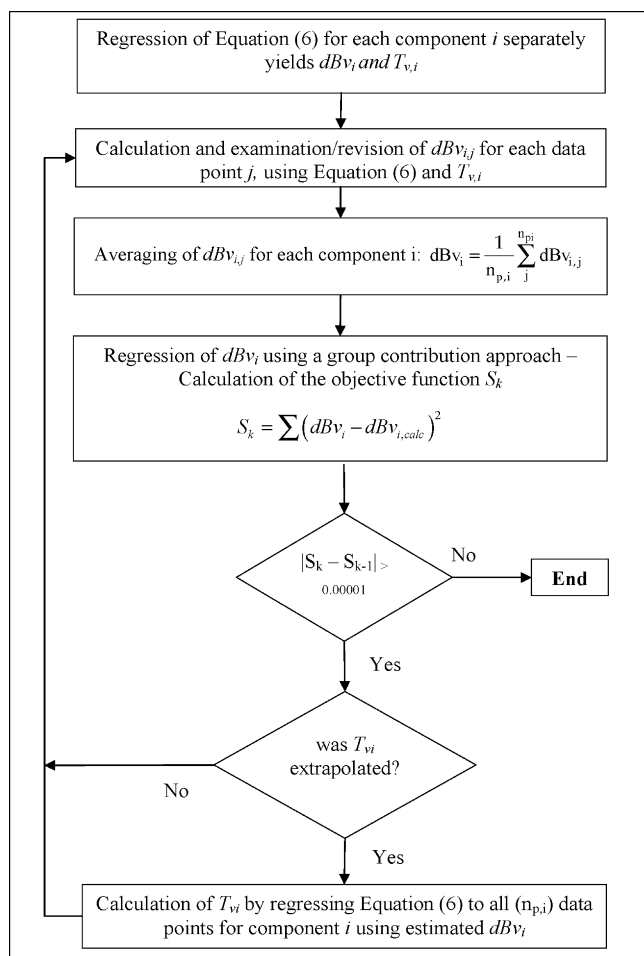


Fig. 1. Flow diagram of the group contribution method development.

3. Development of the new method and results

3.1. Model equations, groups and parameters

It became apparent that for a better representation of liquid viscosities, a slight curvature would have to be modelled. As for the case of vapour pressures, the Vogel equation (3) (having the same form as the Antoine equation) was used and the third parameter was linked to the reference temperature as a convenient reference point:

$$C \approx \frac{T_v}{s} \quad (5)$$

with T_v —viscosity reference temperature (K).

In the case of vapour pressure estimation, a value of $s=8$ was used. From the numerous investigations and optimisations of the viscosity model, a value of $s=16$ produced the most accurate results. Rearranging the Vogel equation with the third parameter from Eq. (5) yields the following expression:

$$\ln \left(\frac{\eta}{1.3cP} \right) = -dBv \left(\frac{T - T_v}{T - (T_v/16)} \right) \quad (6)$$

Eq. (6) is the final model employed in this work to estimate the liquid viscosity.

The procedure used for the development of the method is given in Fig. 1. In the first step, a regression of the experimental data using Eq. (6) was performed in order to determine a reliable reference temperature. The reference viscosity was arbitrarily set to a

value of $\eta_{ref} = 1.3$ cP, which is close to the mean value of all available experimental data.

Using the reference temperatures derived for each component, dBv values were calculated for each data point and carefully examined and revised. The reliable dBv values for each component were then averaged and regressed using the group contribution approach:

$$dBv = \frac{\sum_i^M N_i C (dBv)_i}{n^a + b} + c \quad (7)$$

The values of a , b and c were optimised by non-linear regression, minimising the sum of squared errors (RMSD). n is the number of atoms in the molecule, except hydrogen.

In cases where extrapolation had been required for T_v (i.e. T_v lay outside the temperature range of the experimental data for the respective component), the reference temperatures were again derived from the experimental data by regression using Eq. (6). In this case, the fitting routine used the dBv value from group contribution estimation as a starting value and only allowed optimisation of this value within a small numeric range.

Using the new reference temperatures, new dBv values could be calculated for each data point. Averaging and regression of group contribution parameters led to an improved estimation method. This procedure was repeated until no significant change in the group contributions was observed between consecutive iterations. The final values of the constants for Eq. (7) are:

$$\begin{aligned} a &= -2.5635 \\ b &= 0.0685 \\ c &= 3.7777 \end{aligned}$$

In the last step, a group contribution method for the estimation of the reference temperature was developed.

Out of the many functional relationships evaluated to calculate the reference temperature from the sum of group contributions, one proved especially successful. It employs the sum of group contributions, $\sum_i^M N_i C (T_v)_i$, the normal boiling temperature T_b , and n :

$$T_v = aT_b^{0.5} + \frac{(\sum_i^M N_i C (T_v)_i)^b}{n^c + d} - e \quad (8)$$

with $a=21.8444 \text{ K}^{1/2}$, $b=0.9315$, $c=0.6577$, $d=4.9259$, and $e=231.1361 \text{ K}$.

In Eq. (8) the first two terms show a strong intercorrelation. For this reason, Eq. (9) was regressed first to obtain the parameter a and the exponent b for the normal boiling point term.

$$T_v = aT_b^b + c \quad (9)$$

These values were then set constant in the regression of the further parameters in Eq. (8). After successful regression of these constants and group contributions a simultaneous regression of all parameters was performed leading to another slight improvement.

When estimating liquid viscosities, estimation of the reference temperature should only be used if no reliable data point for a component is available. Otherwise the reference temperature can be calculated from the experimental data and the estimated value of dBv . Estimation of dBv is generally more reliable than the estimation of T_v .

The new method employs the same fragmentation scheme as the method for the estimation of vapour pressure [4,5]. The lists of structural groups for the new method, second order corrections and interacting groups are given in Tables 2–4, respectively. Table 2 contains a priority value for each structural group. For a correct fragmentation of the molecular structure into groups, the different groups need to be matched in the order of their respective priority

Table 2

Group definitions (ID—identification number; PR—priority).

Group	Description	Name	ID, PR	Occurs, e.g. in
Periodic group ^a 17				
Fluorine				
F–	F– connected to non-aromatic C or Si	F–(C,Si)	19, 87	2-Fluoropropane, trimethylfluorosilane
	F– connected to C or Si with at least one F or Cl neighbor and one other atom	F–((C,Si)–([F,Cl]))–a	22, 84	1-Chloro-1,2,2,2-tetrafluoroethane[R124], difluoromethylsilane
	F– connected to C or Si already substituted with at least one F and two other atoms	F–((C,Si)–([F]))–b	21, 81	1,1,1-Trifluoroethane, 2,2,3,3-tetrafluoropropionic acid
	F– connected to C or Si already substituted with at least one Cl and two other atoms	F–((C,Si)–(Cl))–b	102, 82	Trichlorofluoromethane[R11], 1,1-dichloro-1-fluoroethane [R141B]
	F– connected to C or Si already substituted with two F or Cl	F–((C,Si)–([F,Cl] ₂))	23, 83	1,1,1-Trifluorotoluene, 2,2,2-trifluoroethanol, trifluoroacetic acid
	F– connected to an aromatic C	F–(C(a))	24, 86	Fluorobenzene, 4-fluoroaniline
	F– on a C=C (vinylfluoride)	F–C=C<	20, 85	Vinyl fluoride, trifluoroethene, perfluoropropylene
Chlorine				
Cl–	Cl– connected to C or Si not already substituted with F or Cl	Cl–(C,Si)	25, 71	Butyl chloride, 2-chloroethanol, chloroacetic acid
	Cl– connected to C or Si already substituted with one F or Cl	Cl–((C,Si)–([F,Cl]))	26, 70	Dichloromethane, dichloroacetic acid, dichlorosilane
	Cl– connected to C or Si already substituted with at least two F or Cl	Cl–((C,Si)–([F,Cl] ₂))	27, 68	Ethyl trichloroacetate, trichloroacetonitrile
	Cl– connected to an aromatic C	Cl–(C(a))	28, 72	Chlorobenzene
	Cl– on a C=C (vinylchloride)	Cl–C=C<	29, 69	Vinyl chloride
COCl–	COCl– connected to C (acid chloride)	COCl–	77, 18	Acetyl chloride, phenylacetic acid chloride
Bromine				
Br–	Br– connected to a non-aromatic C or Si	Br–(C,Si(na))	30, 65	Ethyl bromide, bromoacetone
	Br– connected to an aromatic C	Br–(C(a))	31, 66	Bromobenzene
Iodine				
I–	I– connected to C or Si	I–(C,Si)	32, 63	Ethyl iodide, 2-iodotoluene
Periodic group 16				
Oxygen				
–OH	–OH for aliphatic chains with less than five C (cannot be connected to aromatic groups)	–OH (<C ₅) (z)	36, 92	Ethanol, propanediol
	–OH connected to C or Si substituted with one C or Si in an at least five C or Si containing chain (primary alkanols)	–OH (>C ₄) (z)	35, 88	1-Nonanol, tetrahydrofurfuryl alcohol, ethylene cyanohydrin
	–OH connected to a C or Si substituted with two C or Si in at least three C or Si containing chain (secondary alkanols)	HO–((C,Si) ₂ H–(C,Si)–(C,Si)–) (z)	34, 90	2-Butanol, cycloheptanol
	–OH connected to C which has four non-hydrogen neighbors (tertiary alkanols)	–OH tert	33, 91	Tert-butanol, diacetone alcohol
	–OH connected to an aromatic C (phenols)	HO–(C(a)) (z)	37, 89	Phenol, methyl salicylate
–O–	–O– connected to two neighbors which are each either C or Si (ethers)	(C,Si)–O–(C,Si) (z)	38, 94	Diethyl ether, 1,4-dioxane
	–O– in an aromatic ring with aromatic C as neighbors	(C(a))–O(a)–(C(a)) (z)	65, 93	Furan, furfural
–CHO	CHO– connected to non-aromatic C (aldehydes)	CHO–(Cna) (z)	52, 53	Acetaldehyde, pentanediol
	CHO– connected to aromatic C (aldehydes)	CHO–(C(a)) (z)	90, 52	Furfural, benzaldehyde
>C=O	–CO– connected to two non-aromatic C (ketones)	O=C<(Cna) ₂ (z)	51, 4	Acetone, methyl cyclopropyl ketone
	–CO– connected to two C with at least one aromatic C (ketones)	(O=C<(C(Ca))) (z)	92, 54	Acetophenone, benzophenone
	–CO connected to N	>N–(C=O)	109, 39	Methyl thioacetate
	–CO connected to two N (urea)	>N–(C=O)–N<	100, 3	Urea-1,1,3,3-tetramethyl
	1,2-Diketone (Do not fragment)	O=C–C=O	118, 1	2,3-Butandione
O=C(–O–) ₂	Non-cyclic carbonate diester	O=C(–O–) ₂	79, 14	Dimethyl carbonate
COOH–	–COOH connected to C (carboxylic acid)	COOH–(C) (z)	44, 23	Acetic acid
–COO–	HCOO– connected to C (formic acid ester)	HCOO–(C) (z)	46, 26	Ethyl formate, phenyl formate
	–COO– connected to two C (ester) in a chain	(C)–COO–(C) (z)	45, 24	Ethyl acetate, vinyl acetate
	–COO– in a ring, C is connected to C (lactone)	–C(r)OO– (z)	47, 25	ϵ -Caprolactone, crotonolactone
–OCOO–	–CO connected to two O (carbonates)	–OCOO–	103, 33	Propylene carbonate, 1,3-dioxolan-2-one
–OCON<	–CO connected to O and N (carbamate)	–OCON<	99, 2	Trimethylsilyl methylcarbamate
>(OC ₂)<	>(OC ₂)< (epoxide)	>(OC ₂)< (z)	39, 50	Propylene oxide

Table 2 (Continued)

Group	Description	Name	ID, PR	Occurs, e.g. in
–CO–O–CO–	Anhydride connected to two C	–C=O–O–C=O–	76, 11	Acetic anhydride, butyric anhydride
	Cyclic anhydride connected to two C connected by a double bond or aromatic bond	(–C=O–O–C=O–)r	96, 10	Maleic anhydride, phthalic anhydride
–O–O–	Peroxide	–O–O–	94, 31	Di-tert-butylperoxide
Sulphur				
–S–S–	–S–S– (disulfide) connected to two C	(C)–S–S–(C)	55, 51	Dimethyldisulfide, 1,2-dicyclopentyl-1,2-disulfide
–SH	–SH connected to C (thiols, mercaptanes)	SH–(C) (z)	53, 73	1-Propanethiol
–S–	–S– connected to two C (thioether)	(C)–S–(C) (z)	54, 74	Methyl ethyl sulfide
	–S– in an aromatic ring (aromatic thioether)	–S(a)– (z)	56, 75	Thiazole, thiophene
–SO ₂ –	Non-cyclic sulfone connected to two C (sulfone)	(C)–SO ₂ –(C)	82, 17	Sulfolane, divinylsulfone
>SO ₄	Sulfates	>SO ₄	104, 34	Dimethyl sulfate
–SO ₂ N<	–S(=O) ₂ connected to N (sulfonamide)	–SO ₂ N<	105, 35	N,N-diethylmethanesulfonamide
>S=O	Sulfoxide	>S=O	107, 37	1,4-Thioxane-S-oxide, tetramethylene sulfoxide
SCN–	SCN– (isothiocyanate) connected to C	SCN–(C)	81, 19	Allyl isothiocyanate
Selenium				
>Se<	>Se< connected to at least one C or Si	>Se<	116, 46	Dimethyl selenide
Periodic group 15				
Nitrogen				
NH ₂ –	NH ₂ – connected to either C or Si (primary amine)	NH ₂ –(C,Si) (z)	40, 96	Hexylamine, ethylenediamine
	NH ₂ – connected to an aromatic C (aromatic primary amine)	NH ₂ –(C(a)) (z)	41, 95	Aniline, benzidine
–NH–	–NH– connected to two C or Si neighbors (secondary amine)	(C,Si)–NH–(C,Si) (z)	42, 100	Diethylamine, diallyl amine
	–NH– connected to two C or Si neighbors in a ring (cyclic secondary amine)	(C,Si)r–NH–(C(a),Si)r (z)	97, 99	Morpholine, pyrrolidine
>N<	>N< connected to three C or Si neighbors (tertiary amine)	(C,Si) ₂ >N–(C,Si)	43, 101	N,N-dimethylaniline, nicotine
	>N< connected to four C or Si (quaternary amine)	(C,Si) ₂ >N<(C,Si) ₂	101, 32	N,N,N,N-tetramethylmethylenediamine
=N–	Double bonded amine connected to at least a C or Si	(C,Si)=N–	91, 102	Acetonin
–N–	Aromatic –N– in a 5-membered ring, free electron pair	=N(a)–(r5) (z)	66, 98	Piperidine, thiazole
=N–	Aromatic =N– in a 6-membered ring	=N(a)–(r6) (z)	67, 97	Pyridine, nicotine
C≡N–	–C≡N (cyano-group) connected to C (cyanide)	(C)–C≡N (z)	57, 55	Acetonitrile, 2,2'-dicyano diethyl sulfide
	–C≡N (cyano-group) connected to N (cyanamide)	(N)–C≡N	111, 41	Dimethylcyanamide
	–C≡N (cyano-group) connected to S (thiocyanate)	(S)–C≡N	108, 38	Methyl thiocyanate
CNCNC–r	Imadizole (aromatic 5-ring)	..=CNC=NC=..	106, 36	1-Methyl-1-imadizole
–CONH<	–CONH ₂ (amide)	–CONH ₂	50, 27	Acetamide
	–CONH– (monosubstituted amide)	–CONH–	49, 48	N-methylformamide, 6-caprolactam
	–CON< (disubstituted amide)	–CON<	48, 49	N,N-dimethylformamide (DMF)
OCN–	OCN– connected to C or Si (isocyanate)	OCN– (z)	80, 28	Butylisocyanate, hexamethylene diisocyanate
ONC–	ONC– (oxime)	ONC–	75, 29	Methyl ethyl ketoxime
–ON=	–ON= connected to C or Si (isoxazole)	–ON=(C,Si)	115, 45	Isoxazole, 5-phenyl isoxazole
NO ₂ –	Nitrites (esters of nitrous acid)	O=N–O–(C)	74, 22	Ethyl nitrite, nitrous acid methyl ester
	NO ₂ – connected to aliphatic C	NO ₂ –(C)	68, 20	1-Nitropropane
	NO ₂ – connected to aromatic C	NO ₂ –(C(a)) (z)	69, 21	Nitrobenzene
NO ₃ –	Nitrate (esters of nitric acid)	NO ₃ –	72, 13	N-butylnitrate, 1,2-propanediol dinitrate
Phosphorous				
>P(O–) ₃	Phosphate triester	PO(O–) ₃	73, 9	Triethyl phosphate, tris-(2,4-dimethylphenyl) phosphate
>P<	Phosphorus connected to at least one C or S (phosphine, phosphane)	>P–(C,Si)	113, 43	Triphenylphosphine, triethylphosphane
Arsine				
AsCl ₂ –	AsCl ₂ connected to C	AsCl ₂ –	84, 16	Ethylarsenic dichloride

Table 2 (Continued)

Group	Description	Name	ID, PR	Occurs, e.g. in
Periodic group 14				
Carbon				
-CH ₃	CH ₃ – not connected to either N, O, F or Cl	CH ₃ –(ne)	1, 105	Decane
	CH ₃ – connected to either N, O, F or Cl	CH ₃ –(e)	2, 103	Dimethoxymethane, methyl butyl ether
-CH ₂ –	CH ₃ – connected to an aromatic atom (not necessarily C)	CH ₃ –(a)	3, 104	Toluene, p-methyl-styrene
	-CH ₂ – in a chain	-C(c)H ₂ –	4, 112	Butane
>CH–	-CH ₂ – in a ring	-C(r)H ₂ –	9, 113	Cyclopentane
	>CH– in a chain	>C(c)H–	5, 119	2-Methylpentane
>C<	>CH– in a ring	>C(r)H–	10, 118	Methylcyclohexane
	>C< in a chain	>C(c)<	6, 121	Neopentane
	>C< in a chain connected to at least one aromatic carbon	>C(c)<(a)	8, 109	Ethylbenzene, diphenylmethane
	>C< in a chain connected to at least one F, Cl, N or O	>C(c)<(e)	7, 108	Ethanol
	>C< in a ring	>C(r)<	11, 120	Beta-pinene
	>C< in a ring connected to at least one aromatic carbon	>C(r)<(Ca)	14, 107	Indene, 2-methyl tetralin
	>C< in a ring connected to, at least one N or O which are not part of the ring, or one Cl or F	>C(r)<(e,c)	12, 110	Cyclopentanol, menthol
	>C< in a ring connected to at least one N or O which are part of the ring	>C(r)<(e,r)	13, 111	Morpholine, nicotine
=C(a)<	Aromatic =CH–	=C(a)H–	15, 106	Benzene
	Aromatic =C< not connected to either O,N,Cl or F	=C(a)<(ne)	16, 117	Ethylbenzene, benzaldehyde
	Aromatic =C< with three aromatic neighbors and three aromatic bonds	(a)=C(a)< ₂ (a)	18, 115	Naphthalene, quinoline
	Aromatic =C< connected to either O,N,Cl or F	=C(a)<(e)	17, 114	Aniline, phenol
	Aromatic =C< with three aromatic neighbors and two aromatic bonds (aliphatic bridge bond between aromatic rings)	C(a)=C(a)<C ₂ (a) (bridge)	214, 116	Biphenyl, m-terphenyl
>C=C<	H ₂ C=C< (1-ene)	H ₂ C(c)=C<	61, 57	1-Hexene
	>C=C< (both C have at least one non-H neighbor)	>C(c)=C(c)<	58, 62	2-Heptene, mesityl oxide
	Non-cyclic >C=C< connected to at least one aromatic C	>C(c)=C(c)<(C(a))	59, 59	Isosafrole, cinnamic alcohol
	Cyclic >C=C<	>C(r)=C(r)<	62, 60	Cyclopentadiene
	Non-cyclic >C=C< substituted with at least one F, Cl, N or O	-(e)C(c)=C(c)<	60, 58	trans-1,2-Dichloroethylene, perfluoroisoprene
-C≡C–	HC≡C– (1-ine)	HC≡C–	64, 56	1-Heptyne
	-C≡C– with two non-H neighbors	-C≡C–	63, 61	2-Octyne
>C=C=C<	Two cumulated double bonds	>C=C=C<	87, 5	1,2-Butadiene, dimethyl allene
>C=C–C=C<	Two conjugated double bonds in a ring	>C=C–C=C<	88, 6	Cyclopentadiene, abietic acid
>C=C–C=C<	Two conjugated double bonds in a chain	>C=C–C=C<	89, 7	Isoprene, 1,3-hexadiene
-C≡C–C≡C–	Two conjugated triple bonds	-C≡C–C≡C–	95, 8	2,4-Hexadiyne
Silicon				
>Si<	>Si<	>Si<	70, 80	Butylsilane
	>Si< attached to no carbon or hydrogen	>Si<(C,H) ₀	216, 76	Tetrachlorosilane, tetramethoxysilane
	>Si< attached to one carbon or hydrogen	>Si<(C,H) ₁	215, 77	Trichlorosilane, methyltrichlorosilane
	>Si< attached to two carbon or hydrogen	>Si<(C,H) ₂	93, 78	Dichlorodimethylsilane, dichlorodiethylsilane
	>Si< attached to three carbon or hydrogen	>Si<(C,H) ₃	71, 79	Hexamethyl disiloxane
Germanium				
>Ge<	>Ge< connected to four carbons	(C) ₂ >Ge<(C) ₂	86, 67	Tetramethylgermane
GeCl ₃ –	GeCl ₃ – connected to carbons	GeCl ₃ –(C)	85, 12	Fluorodimethylsilyl (trichlorogermanyl)methane
Tin				
>Sn<	>Sn< connected to four carbons	(C) ₂ >Sn<(C) ₂	83, 64	Tetramethylstannane
Periodic group 13				
Boron				
B(O–) ₃	Non-cyclic boric acid ester	>B(O–) ₃	78, 15	Triethyl borate
Aluminum				
>Al<	>Al< connected to at least one C or Si	>Al<(C, Si)	117, 47	Triethylaluminum

Abbreviations: (e) very electronegative neighbors (N, O, F, Cl); (ne) not very electronegative neighbors (not N, O, F, Cl); (na) non-aromatic atom or neighbor; (a) aromatic atom or neighbor; (c) atom or neighbor is part of a chain; (r) atom or neighbor is part of a ring; (z) groups considered to be non-additive.

^a Periodic group number of the most significant element in the structural group. Periodic group numbers are used as defined in the IUPAC version of the periodic table of elements.

Table 3

Second-order groups and corrections.

Name contribution (K)	Description	ID	Example(s)
C=C–C=O	–C=O connected to sp ² carbon	134	Benzaldehyde furfural
(C=O)–C([F,Cl] _{2,3})	Carbonyl connected to carbon with two or more halogens	119	Dichloroacetyl chloride
(C=O)–C([F,Cl] _{2,3}) ₂	Carbonyl connected to two carbon with two or more halogens each	120	Perfluoro-2-propanone
C–[F,Cl] ₃	Carbon with three halogens	121	1,1,1-Trifluorotoluene
(C) ₂ –C–[F,Cl] ₂	Secondary carbon with two halogens	122	2,2-Dichloropropane
No hydrogen	Component has no hydrogen	123	Perfluoro compounds
One hydrogen	Component has one hydrogen	124	Nonafluorobutane
3/4 ring	A three or four-membered non-aromatic ring	125	Cyclobutene
5 ring	A five-membered non-aromatic ring	126	Cyclopentane
Ortho-pair(s)	Ortho position—counted only once and only if there are no meta or para pairs	127	o-Xylene
Meta-pair(s)	Meta position—counted only once and only if there are no para or ortho pairs	128	m-Xylene
Para-pair(s)	Para position—counted only once and only if there are no meta or ortho pairs	129	p-Xylene
((C=C)C)C–CC ₃	Carbon–carbon bond with four single bonded and one double bonded carbon neighbor	130	tert-Butylbenzene
C ₂ C–CC ₂	Carbon–carbon bond with four carbon neighbors, two on each side	131	Bicyclohexyl
C ₃ C–CC ₂	Carbon–carbon bond with five carbon neighbors	132	Ethyl bornyl ether
C ₃ C–CC ₃	Carbon–carbon bond with six carbon neighbors	133	2,2,3,3-Tetramethylbutane
Si < (F, Cl, Br, I)	A silicon attached to a halogen atom	217	Trichloroethylsilane

Table 4

Groups considered to be non-additive (group-ID(s) given in paranthesis).

Group Abbr.	Group name (group ID(s))	Group Abbr.	Group name (group ID(s))
OH	Alkanol (–OH) (34, 35, 36)	Ats	Aromatic sulphur (–S(a)–) (56)
OH(a)	Phenol (–OH(a)) (37)	SH	Thiol (–SH) (53)
COOH	Carboxylic acid (–COOH) (44)	NH ₂	Primary amine (–NH ₂) (40, 41)
EtherO	Ether (–O–) (38)	NH	Secondary amine (>NH) (42, 97)
Epox	Epoxide (>(OC ₂)<) (39)	OCN	Isocyanate (–OCN) (80)
Ester	Ester (–COOC–) (45, 46, 47)	CN	Cyanide (–CN) (57)
Ketone	Ketone (–CO–) (51, 92)	Nitro	Nitrate (69)
Alde	Aldehyde (–CHO) (52, 90)	AN5	Aromatic N in 5-ring (=N(a)–(r5)) (66)
AO	Aromatic oxygen (–O(a)–) (65)	AN6	Aromatic N in 6-ring (=N(a)–(r6)) (67)
Teth	Sulfide (thioether) (–S(na)–) (54)		

number. In general, more specific or larger groups are assigned a smaller priority number and are therefore matched (detected) first.

The final group contribution values are given in Tables 5–7 for estimation of dB_v and Tables 8–10 for the estimation of the viscosity reference temperature T_v . Tables 5 and 8 list contributions of individual structural groups, second order corrections can be found in Tables 6 and 9 and group interaction contributions in Tables 7 and 10, respectively. In each of these tables, also the absolute mean deviation,² relative mean deviation³ and standard deviation of the estimated parameter (dB_v and T_v , respectively) are given. These were calculated for all components in the training set that contain the, respectively, group, correction or interaction contribution.

For a number of the groups listed in Table 2, no parameters are available and liquid viscosity estimation is not possible. The groups are nevertheless reported. In case data or reasonable estimates for other components containing these groups are available to the user, it is possible to calculate or by regression determine the missing contribution. If the groups would have been discarded from the method, it might still have been possible to perform property estimation using simpler (lower priority) groups, but this could possibly lead to significantly larger errors. For interacting groups, not every interaction is of similar importance. Ignoring specific group interactions will lead to similar deviations as demonstrated in the case of normal boiling point estimation [2].

3.2. Hydrocarbon compounds

As all literature group contribution methods for the estimation of liquid viscosities evaluated in this paper (see Table 1) have proven to be of low reliability, they did not constitute a good reference for the performance of the new method. For this reason, estimation results are compared to the results of correlative models such as the Andrade and Vogel equations. In many cases, the deviation observed for these models represents the statistical scatter (“noise”) of the data. In other cases they may fit questionable data with high accuracy even though the much higher deviating results of the estimation method may be more realistic.

Vogel equation parameters were to a great part already available in the DDB and only several additional parameter sets had to be regressed. Andrade parameters were regressed for all components in this work.

In order to be able to judge the quality of the group contribution parameter regression, relative mean deviations in logarithmic liquid viscosity are reported for the different saturated vapour pressure ranges, whereby the vapour pressure was estimated by group contribution [4]. The vapour pressure range was chosen as it represents a certain relation between thermal energy and intermolecular forces, and at the same time an approximate viscosity range.

Since there were only few data points at high vapour pressures (>500 kPa), this range has been merged into the moderate pressure range (MP). The vapour pressure ranges are defined as:

- ELP vapour pressure below 0.01 kPa;
- LP vapour pressure between 0.01 and 10 kPa;
- MP vapour pressure larger than 10 kPa.

² Absolute mean deviation: $(1/n_p) \sum |x_{\text{calc}} - x_{\text{exp}}|$.

³ Relative mean deviation: $(1/n_p) \sum |(x_{\text{calc}} - x_{\text{exp}})/x_{\text{exp}}|$.

Table 5Saturated liquid viscosity curve slope (dBv) group contributions, number of components used for regressing these values and deviations in dBv for these components.

Group ID	Group contribution, $dBv_i (\times 10^3)$	Number of components	Absolute mean deviation	Relative mean deviation (%)	Standard deviation
1	13.9133	520	0.2	3.2	0.3
2	11.7002	70	0.2	3.2	0.3
3	−11.0660	46	0.2	3.3	0.3
4	2.1727	344	0.2	3.0	0.3
5	4.5878	90	0.2	3.0	0.4
6	37.0296	22	0.3	5.1	0.5
7	21.3473	331	0.2	3.4	0.4
8	5.9452	28	0.4	7.6	0.5
9	10.8799	73	0.2	3.5	0.4
10	−7.2202	37	0.2	3.8	0.4
11	142.1976	3	0.5	7.6	0.6
12	61.0811	12	0.4	4.8	0.7
13	28.7351	19	0.1	2.5	0.2
14	−12.3456	3	0.2	5.0	0.2
15	−2.7840	184	0.2	4.3	0.4
16	45.9403	114	0.2	4.4	0.4
17	80.5124	89	0.2	4.3	0.4
18	37.4124	13	0.2	3.9	0.3
19	5.1640	8	0.1	3.6	0.2
21	2.8323	38	0.2	3.3	0.2
22	0.7129	15	0.1	3.3	0.2
23	−36.3189	2	0.0	0.6	0.0
24	−61.9434	6	0.0	0.5	0.0
25	4.7579	32	0.2	3.3	0.3
26	5.8228	18	0.1	3.0	0.2
27	4.6555	30	0.1	2.1	0.1
28	−67.3989	23	0.3	5.4	0.5
29	−9.6209	5	0.0	0.6	0.0
30	0.5164	29	0.1	3.7	0.2
31	−18.1984	6	0.1	3.8	0.2
32	−17.3110	15	0.2	3.9	0.3
33	336.8834	10	0.7	7.6	1.0
34	365.8067	38	0.4	4.5	0.6
35	249.0118	40	0.4	5.2	0.6
36	218.8000	23	0.3	4.4	0.4
37	160.8315	16	0.3	4.9	0.4
38	−35.3055	99	0.2	3.8	0.3
39	85.3693	3	0.0	0.2	0.0
40	58.9131	20	0.2	2.9	0.3
41	44.0698	16	0.4	6.8	0.6
42	13.6479	10	0.2	2.8	0.2
43	−58.7354	10	0.3	4.5	0.4
44	54.7891	26	0.2	3.8	0.3
45	17.6757	69	0.2	2.9	0.3
46	0.4267	9	0.1	1.2	0.1
47	47.6109	1			
48	12.6717	6	0.1	2.7	0.2
49	129.8293	3	0.5	8.7	0.6
50	202.2864	3	0.1	1.4	0.1
51	24.2524	24	0.1	2.7	0.2
52	18.4961	7	0.0	0.3	0.0
53	30.5022	12	0.1	2.5	0.2
54	−0.0276	13	0.0	0.8	0.0
55	−13.4614	2	0.0	0.1	0.0
56	18.5507	3	0.0	0.5	0.0
57	23.1459	19	0.1	2.2	0.2
58	9.5809	8	0.1	2.0	0.2
59	152.2693	2	0.2	2.4	0.2
60	18.5983	4	0.0	0.6	0.0
61	21.4560	29	0.1	1.4	0.2
62	19.7836	5	0.2	3.6	0.2
63	−165.0071	1			
64	13.3585	3	0.4	8.1	0.5
65	42.7958	3	0.0	0.0	0.0
66	151.9493	2	0.2	3.1	0.2
67	52.5900	18	0.2	3.9	0.3
68	−34.3948	6	0.1	1.7	0.1
69	−6.5626	11	0.3	6.5	0.5
70	−25.5950	2	0.0	0.2	0.0
71	−28.3943	10	0.2	4.5	0.2
72	−30.6156	2	0.0	0.0	0.0
73	45.9972	4	0.1	1.6	0.1
74	−7.3298	5	0.0	0.2	0.0
75	369.8367	1			

Table 5 (Continued)

Group ID	Group contribution, dBv_i ($\times 10^3$)	Number of components	Absolute mean deviation	Relative mean deviation (%)	Standard deviation
76	16.3525	4	0.1	2.6	0.2
77	−2.6553	6	0.0	1.0	0.1
78	−61.2368	6	0.1	1.1	0.1
79	−7.5067	2	0.0	0.9	0.0
80	4.1408	8	0.1	1.9	0.1
81	−46.5613	2	0.0	1.1	0.0
82	122.6902	3	0.0	0.7	0.0
86	−70.9713	1			
88	29.0985	1			
89	125.0861	2	0.0	0.2	0.0
90	−14.2823	3	0.0	0.0	0.0
92	−8.5352	4	0.2	4.0	0.2
93	9.9037	13	0.2	4.8	0.3
96	102.0816	2	0.1	2.7	0.1
97	74.0520	3	0.0	0.4	0.0
98	43.6079	5	0.2	3.7	0.3
100	54.4769	1			
102	5.7765	3	0.0	0.5	0.0
103	95.6531	1			
104	56.9133	2	0.3	5.9	0.3
105	64.7133	3	0.1	1.0	0.1
107	22.9969	1			
108	23.2473	2	0.1	1.4	0.1
110	−45.7263	4	0.2	4.1	0.2
118	(do not estimate) ^a	–			
214	37.5669	5	0.1	2.6	0.2
215	64.6600	3	0.0	0.6	0.0
216	68.4952	4	0.0	0.9	0.0

^a Several groups, corrections or interaction groups may cover situations where estimations would lead to large errors and our methods are explicitly not to be applied to these cases. Group 118 ((−C=O)−(C=O)−, Table 2 under “>C=O”) has the highest possible priority (1) and in case it is found the methods are explicitly not applicable!

Table 6

Saturated liquid viscosity curve slope (dBv) second-order contributions, number of components used for regressing these values and deviations in dBv for these components.

Group ID	Group contribution, dBv_i ($\times 10^3$)	Number of components	Absolute mean deviation	Relative mean deviation (%)	Standard deviation
119	0.3041	7	0.1	2.7	0.2
121	−6.1420	48	0.1	3.4	0.2
122	−26.4635	11	0.2	3.6	0.2
123	−14.9636	30	0.1	3.4	0.2
124	−25.9017	17	0.1	2.9	0.1
125	−57.3789	7	0.0	0.9	0.0
126	−21.2204	32	0.1	2.3	0.2
127	−20.1917	53	0.2	3.9	0.4
128	−34.5860	33	0.2	4.2	0.4
130	−110.7391	3	0.6	9.0	0.7
131	2.4859	18	0.3	4.1	0.5
132	−59.3670	4	0.0	1.0	0.0
134	13.1413	26	0.1	2.7	0.2
217	−76.1631	9	0.0	1.3	0.1

The development of the proposed group contribution model for the estimation of liquid viscosities started with the regression of viscosities of the n-alkanes. In the first regression, data were verified to allow the model development to start from a ‘clean’ set of data. Subsequent regressions revealed an excellent representation of the dBv parameter by group contribution. A plot of the liquid viscosity estimation results for n-alkanes is presented in Fig. 2 together with experimental data from the DDB. For all compounds, estimations from this work are in excellent agreement with the experimental data. The estimation of the slope shows no variance with increasing molecular weight and can be assumed to extrapolate correctly with respect to chemical constitution. The close proximity of the higher molecular weight curves also suggest that the change in viscosity between consecutive members in the series is decreasing.

Due to the similarities between viscosity and vapour pressure which have been discussed above; as in the case of vapour pressure estimation, the mean temperature difference between the experimental value and estimation for the same experimental viscosity is a convenient measure of deviation. It can therefore be expected that

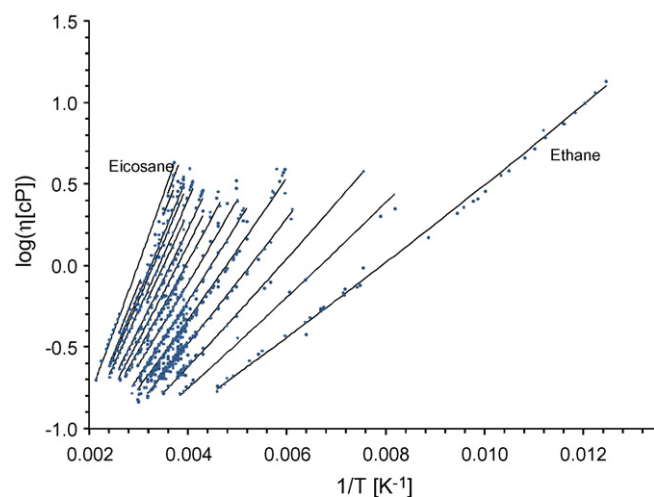


Fig. 2. Liquid viscosity estimation results for n-alkanes (ethane to eicosane) together with experimental data from the DDB using adjusted reference temperatures.

Table 7
Saturated liquid viscosity curve slope ($d\ln v$) group interaction contributions, interacting groups, number of components used for regressing these values and deviations in $d\ln v$ for these components.

Group ID	Interacting groups ^a	Group contribution, $d\ln v_i$ ($\times 10^3$)	Number of components	Absolute mean deviation	Relative mean deviation (%)	Standard deviation
135	OH–OH	–112.4939	19	0.1	1.2	0.2
136	OH–NH ₂	1031.5920	3	0.2	1.7	0.2
137	OH–NH	853.2318	2	0.1	0.6	0.1
140	OH–EtherO	–423.9834	15	0.4	7.2	0.6
145	OH–CN	–683.0189	1			
146	OH–AO	–557.5079	1			
148	OH(a)–OH(a)	–1186.0500	1			
151	OH(a)–EtherO	–333.5638	4	0.1	1.1	0.1
155	OH(a)–Nitro	–878.0615	3	0.3	5.3	0.3
157	NH ₂ –NH ₂	135.3183	3	0.6	9.0	0.7
159	NH ₂ –EtherO	219.9701	5	0.2	3.4	0.3
166	NH–EtherO	–134.4625	1			
178	EtherO–EtherO	132.0275	40	0.2	3.9	0.3
180	EtherO–Ester	44.8702	1			
181	EtherO–Ketone	–219.5265	2	0.1	1.9	0.1
182	EtherO–Alde	546.5846	1			
184	EtherO–Nitro	–59.3635	1			
189	Ester–Ester	964.0840	11	0.3	3.2	0.4
190	Ester–Ketone	126.0380	2	0.0	0.2	0.0
192	Ester–CN	539.2401	2	0.1	1.7	0.1
194	Ketone–Ketone	3705.4400 ^b	1			
204	Alde–AO	50.1063	1			
206	Nitro–Nitro	896.3606	1			
209	CN–AN6	–196.6361	1			
218	COOH–NH ₂	(do not estimate) ^c	–			

^a Group abbreviations used as defined in Table 4.

^b Questionable group contribution values.

^c Several groups, corrections or interaction groups may cover situations where estimations would lead to large errors and our methods are explicitly not to be applied to these cases. Components containing both the COOH and NH₂ group (amino-acids) carry two opposite charges which has an extreme effect on their properties.

mean temperature errors for both properties should be of similar magnitude.

Higher deviations in the case of hydrocarbons were observed for cyclohexane and *cis*-decahydronaphthalene. These deviations are not untypical for smaller molecules like cyclohexane consisting solely of one type of structural group, but in this case are the result of the higher rotational symmetry of the molecule as discussed below. *cis*-Decahydronaphthalene shows a very peculiar structure compared to the *trans*-form and the method contains no correction for the *cis*- and *trans*-forms. Nevertheless, even these deviations are well below 20 K.

Detailed results for the different types of hydrocarbons for both the proposed method and the correlative models are presented in Tables 11 and 12, respectively. The proposed method yields a consistent and accurate set of results for the different classes of hydrocarbons and only slightly higher deviations than the direct correlation.

The results reported for the viscosity reference temperature show higher deviations for cyclic alkanes, aromatics and the smaller molecules, generally those with higher rotational symmetry number.

Cyclooctane has a viscosity reference temperature of 327.0 K with a rotational symmetry of 8. Eleven isomers of this compound have an average reference temperature of 264.8 K with a highest rotational symmetry of 2. The difference between the reference temperature for cyclooctane and the isomer with the highest temperature is 33.7 K. Cyclohexane and methyl cyclopentane with a rotational symmetry of 6 and 1, respectively; show a difference of 46.2 K with the former compound having the higher reference temperature.

Higher symmetry seems to decrease the entropy difference between disordered and structured liquid states and thus favours the latter leading to the observed increase in viscosity. Due to difficulties in deriving the symmetry of molecules from

the molecular structure by an automated algorithm, this effect was not included in the current estimation method for liquid viscosity.

3.3. Mono-functional compounds

3.3.1. Oxygen compounds

Results for the different types of alcohol compounds for both the proposed method and the correlative models are presented in Tables 13 and 14, respectively.

Larger deviations in viscosity reference temperature were found for:

- 2-methyl-1-butanol (26.4 K);
- tert-butanol (24.4 K) (higher rotational symmetry);
- 1-pentanol (18.0 K) (first member of the series OH on a chain longer than 4 carbon atoms);
- 1,2-ethanediol (21.0 K) (first member of the alkane diol series).

The results for the different types of oxygen (except alcohol) compounds for the proposed method and the correlative models are presented in Tables 15 and 16, respectively. The proposed method yields a consistent and accurate set of results for the different classes of oxygenated compounds which is in comparable accuracy to the correlative models. Even at low temperatures, there were no exceptionally high deviations.

For the viscosity reference temperature, large deviations were observed for the smaller compounds. These compounds which included acetic acid (23.3 K), propionic acid (43.2 K), butyric acid (32 K), 2-methylpropionic acid (37.8 K), 2-methyltetrahydrofuran (27.9 K) and acetaldehyde (14.7 K) were removed from the regression set. Even with the removal of the above compounds, there were still large deviations for the smaller carboxylic acids up to hexanoic acid. Apart from these compounds, overall there were no large devi-

Table 8Saturated liquid viscosity reference temperature (T_v) group contributions, number of components used for regressing these values and deviations in T_v for these components.

Group ID	Group contribution, $T_{v,i}$	Number of components	Absolute mean deviation (K)	Relative mean deviation (%)	Standard deviation (K)
1	89.0803	509	6.7	2.3	8.8
2	216.0226	68	7.5	2.7	9.9
3	80.9698	46	4.8	1.6	6.2
4	60.3316	339	6.1	2.1	8.1
5	24.2637	88	7.0	2.5	9.2
6	244.4643	22	9.6	3.5	11.0
7	103.4109	327	6.9	2.3	9.5
8	−16.5212	28	7.3	2.1	9.6
9	174.1316	72	8.3	2.8	10.3
10	−37.7584	37	8.6	2.9	10.0
11	252.0190	3	8.8	3.0	10.2
12	251.9299	12	7.3	2.2	8.9
13	330.7100	18	5.4	1.7	7.8
14	294.3323	3	10.7	3.4	13.2
15	113.9028	184	7.2	2.2	10.8
16	−26.6195	114	6.4	2.0	9.6
17	133.5499	89	8.5	2.5	12.1
18	128.4739	13	3.6	1.0	4.1
19	208.3258	8	8.7	3.8	11.7
21	35.2688	38	6.0	2.7	7.5
22	207.3562	15	6.4	2.8	7.0
23	−15.8544	2	2.1	1.3	2.2
24	112.1172	6	11.3	4.6	12.2
25	329.0113	31	6.5	2.4	7.7
26	313.1106	18	7.9	3.0	9.7
27	194.6060	30	8.6	3.5	12.5
28	−8.6247	23	8.4	2.6	10.8
29	182.7067	5	4.0	1.7	4.3
30	456.3713	29	8.4	3.0	10.6
31	391.6060	6	6.7	2.2	7.0
32	499.2149	14	4.7	1.7	6.1
33	1199.4010	9	6.8	2.1	9.3
34	1198.1040	38	7.4	2.1	8.9
35	1078.0840	38	13.7	3.7	17.0
36	1284.7450	22	10.2	2.8	12.6
37	1134.1640	16	14.5	3.8	19.5
38	−34.9892	98	9.3	3.0	13.1
39	612.7222	3	5.3	2.4	5.6
40	458.7425	19	5.8	2.0	7.8
41	705.1250	16	6.2	1.8	8.2
42	159.5146	10	3.0	0.9	4.6
43	−284.4707	10	10.0	2.8	15.2
44	1446.0240	22	14.4	4.0	16.6
45	325.5736	69	5.5	1.8	7.5
46	454.1671	9	3.4	1.4	4.4
47	374.6477	1			
48	289.9690	6	9.3	3.0	10.2
49	1150.8290	2	8.8	2.4	8.8
50	1619.1650	2	1.2	0.3	1.2
51	304.5982	24	6.5	2.4	7.4
52	394.7932	6	6.1	2.6	6.5
53	294.7319	12	7.2	2.9	8.9
54	206.6432	13	8.4	3.2	11.9
55	292.3613	2	2.6	1.0	2.6
56	302.2321	3	1.5	0.6	1.7
57	346.9998	18	5.1	1.9	6.4
58	−23.9801	8	7.1	2.3	9.1
59	238.3242	2	0.6	0.2	0.6
60	137.5408	4	4.6	2.1	4.7
61	74.4489	29	8.5	3.4	10.8
62	304.9257	5	12.3	4.8	13.7
63	−32.4179	1			
64	57.8131	3	8.0	3.0	8.8
65	279.2114	3	0.0	0.0	0.0
66	662.0051	2	1.2	0.4	1.2
67	277.5038	18	3.3	1.0	5.1
68	369.4221	6	4.1	1.6	4.7
69	488.1136	11	15.3	4.1	21.1
70	−10.6146	2	9.8	3.9	10.3
71	−181.7627	9	20.5	6.6	22.7
72	351.0623	2	0.0	0.0	0.0
73	−15.2801	4	17.0	4.6	19.3
74	174.3672	5	1.5	0.7	1.9
75	1098.1570	1			
76	549.1481	4	2.0	0.7	2.4

Table 8 (Continued)

Group ID	Group contribution, $T_{v,i}$	Number of components	Absolute mean deviation (K)	Relative mean deviation (%)	Standard deviation (K)
77	394.5776	6	2.6	1.0	3.3
78	10.3752	6	9.8	4.2	14.0
79	365.8081	2	5.6	2.2	5.6
80	164.8904	8	8.6	3.6	11.0
81	197.1806	2	0.9	0.4	0.9
82	1297.7560	3	6.6	1.6	7.1
86	35.4672	1			
88	495.5141	1			
89	551.9254	2	1.9	0.9	1.9
90	490.7224	3	0.0	0.0	0.0
92	669.0158	4	8.3	2.4	10.5
93	256.5078	13	16.4	5.0	19.4
96	1787.0390	2	12.2	3.2	12.2
97	220.0803	3	1.0	0.4	1.2
98	229.4135	5	4.0	1.2	4.8
100	131.2253	1			
102	53.2507	3	2.1	0.9	2.3
103	288.4140	1			
104	542.6641	2	4.8	1.5	4.8
105	714.0494	3	4.5	1.3	4.7
107	797.2271	1			
108	253.5303	2	2.9	1.1	3.0
110	−237.2545	4	11.1	3.5	11.6
118	(do not estimate) ^a	–			
214	192.1303	5	5.8	1.5	7.4
215	377.7146	3	30.0	12.5	31.9
216	806.8125	4	13.5	4.8	14.7

^a See footnote of Table 5.

ations for all mono-functional oxygen compounds (except alcohols) greater than 17 K.

3.3.2. Nitrogen compounds

Results for the different types of nitrogen compounds for the proposed method and both correlative models are presented in Tables 17 and 18, respectively. There were no relatively large deviations observed, even at low temperatures.

For all nitrogen compounds (including multi-functional compounds), an average absolute error of 1.8% is tabulated, which is satisfactory considering the errors that may arise from the calculation of this point. N-methylformamide (26.9 K), formamide (26.5 K), methylamine (17.6 K) and acetonitrile (13.5 K) which are the first compounds in their respective series were removed from the regression set. For all mono-functional nitrogen compounds, there were no deviations greater than 20 K.

3.3.3. Sulphur compounds

Results for the different types of sulphur compounds for the proposed method and both correlative models are presented in

Tables 19 and 20, respectively. There were no exceptionally large deviations observed from the estimation of the slope for all sulphur compounds over the entire temperature range.

For all mono-functional sulphur compounds, there were no deviations greater than 18 K. Compared to oxygen and nitrogen, sulphur is a weaker hydrogen bonding acceptor. This implies that the influence of the intermolecular force is weaker on smaller sulphur compounds. Thus, no compounds needed to be removed from the regression set.

3.3.4. Halogen compounds

Results for the different types of halogen compounds for the proposed method and both correlative models are presented in Tables 21 and 22, respectively. In contrary to other functional groups, halogen group contributions are assumed to observe simple additivity and no distinction is made between compounds with one or more halogen atoms. There were no exceptionally large deviations observed from the estimation of the slope for halogen compounds over the entire temperature range.

Table 9

Saturated liquid viscosity reference temperature (T_v) second-order contributions, number of components used for regressing these values and deviations in T_v for these components.

Group ID	Group contribution, $T_{v,i}$	Number of components	Absolute mean deviation (K)	Relative mean deviation (%)	Standard deviation (K)
119	−180.3686	7	8.6	2.7	11.0
121	241.8968	48	7.0	2.8	9.3
122	138.6555	11	4.6	1.8	5.6
123	−71.1647	30	8.0	3.1	10.2
124	−115.0418	17	8.7	3.5	11.1
125	−96.7544	7	6.4	2.6	7.3
126	−153.8442	31	7.5	2.6	9.4
127	−22.1041	53	8.4	2.5	12.1
128	24.7835	33	7.7	2.4	10.6
130	224.2439	3	9.1	2.3	10.9
131	24.2539	18	8.3	2.8	10.1
132	137.8708	4	9.6	3.6	11.0
134	−54.1782	26	8.3	2.6	11.1
217	−726.4291	8	19.5	8.1	22.4

Table 10

Saturated liquid viscosity reference temperature (T_v) group interaction contributions, interacting groups, number of components used for regressing these values and deviations in T_v for these components.

Group ID	Interacting groups ^a	Group contribution, $T_{v,i}$	Number of components	Absolute mean deviation (K)	Relative mean deviation (%)	Standard deviation (K)
135	OH–OH	–1313.5690	18	12.9	3.2	16.7
136	OH–NH ₂	–41.9608	3	13.0	3.4	13.3
137	OH–NH	–1868.6060	2	9.8	2.4	9.8
140	OH–EtherO	–643.4378	15	17.2	4.9	20.4
145	OH–CN	–345.7844	1			
146	OH–AO	50.2582	1			
148	OH(a)–OH(a)	–1146.1070	1			
151	OH(a)–EtherO	–229.2406	4	16.3	4.3	17.3
155	OH(a)–Nitro	515.1511	3	34.1	8.6	37.2
157	NH ₂ –NH ₂	86.7249	3	2.9	0.8	3.5
159	NH ₂ –EtherO	–57.1437	5	6.9	1.8	7.5
166	NH–EtherO	54.2025	1			
178	EtherO–EtherO	156.7495	40	11.0	3.4	15.1
180	EtherO–Ester	273.6616	1			
181	EtherO–Ketone	–339.6071	2	1.9	0.5	1.9
182	EtherO–Alde	1050.3190	1			
184	EtherO–Nitro	355.0508	1			
189	Ester–Ester	167.7204	11	10.1	2.7	12.7
190	Ester–Ketone	244.0583	2	1.8	0.6	1.8
192	Ester–CN	334.4856	2	0.3	0.1	0.3
194	Ketone–Ketone	1985.8270 ^b	1			
204	Alde–AO	161.7447	1			
206	Nitro–Nitro	1839.2630	1			
209	CN–AN6	718.1262	1			
218	COOH–NH ₂	(do not estimate) ^c	–			

^a Group abbreviations used as defined in Table 4.

^b Questionable group contribution values.

^c See footnote of Table 7.

Table 11

Viscosity relative mean deviation (%) and reference temperature average mean deviation (K) of the proposed method for the different types of hydrocarbons (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average; (AAD) average absolute deviation (K); (MAPE) mean average percentage error (%).

	NC	RMD (%) in $\ln(\eta)$				AMD in T_v		
		ELP	LP	MP	AV	NC	AAD	MAPE
Hydrocarbons (HC)	147	5.1 ³²⁰	3.0 ¹⁶⁶²	2.7 ⁸⁷⁷	3.1 ²⁸⁵⁹	147	6.3	2.4
Saturated HC	91	4.9 ²⁸⁰	2.5 ⁹³⁰	2.7 ⁶²⁸	3.0 ¹⁸³⁸	91	6.5	2.5
Non-aromatic HC	113	4.8 ²⁸⁹	2.5 ⁹⁸³	2.6 ⁷³⁹	2.8 ²⁰¹¹	113	6.3	2.5
Unsaturated HC	22	1.5 ⁹	2.1 ⁵³	1.6 ¹¹¹	1.7 ¹⁷³	22	5.6	2.7
n-Alkanes	25	4.2 ²⁵¹	2.4 ⁵³⁶	1.7 ²⁷⁷	2.6 ¹⁰⁶⁴	25	4.2	1.5
Alkanes (non-cyclic)	63	4.9 ²⁷⁶	2.7 ⁷¹⁴	3.3 ⁴³⁴	3.3 ¹⁴²⁴	63	5.7	2.3
Alkanes (cyclic)	28	6.7 ⁴	2.0 ²¹⁶	1.4 ¹⁹⁴	1.7 ⁴¹⁴	28	8.3	3.0
Aromatic HC	34	7.9 ³¹	3.6 ⁶⁷⁹	3.5 ¹³⁸	3.8 ⁸⁴⁸	34	6.4	2.0
Fused aromatic HC	8	11.2 ²	2.8 ⁹³	4.2 ⁶⁴	3.5 ¹⁵⁹	8	5.3	1.4
Alkenes HC	18	1.5 ⁹	1.6 ⁴⁵	1.8 ⁸¹	1.7 ¹³⁵	18	6.1	2.9
Alkenes (cyclic HC)	3	–	1.7 ⁷	0.7 ¹³	1.0 ²⁰	3	11.1	5.1
Alkynes HC	2	–	5.6 ⁷	1.5 ⁷	3.6 ¹⁴	2	5.7	2.4

Table 12

Viscosity relative mean deviation (%) of the Andrade and Vogel model for the different types of hydrocarbons (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average.

	RMD (%) in $\ln(\mu)$									
	Andrade					Vogel				
	NC	ELP	LP	MP	AV	NC	ELP	LP	MP	AV
Hydrocarbons (HC)	147	2.6 ³²⁰	1.1 ¹⁶⁶²	1.5 ⁸⁷⁷	1.4 ²⁸⁵⁹	147	4.6 ³²⁰	3.3 ¹⁶⁶²	3.4 ⁸⁷⁷	3.5 ²⁸⁵⁹
Saturated HC	91	2.4 ²⁸⁰	1.0 ⁹³⁰	1.5 ⁶²⁸	1.4 ¹⁸³⁸	91	2.7 ²⁸⁰	3.5 ⁹³⁰	3.4 ⁶²⁸	3.4 ¹⁸³⁸
Non-aromatic HC	113	2.4 ²⁸⁹	1.0 ⁹⁸³	1.5 ⁷³⁹	1.4 ²⁰¹¹	113	2.9 ²⁸⁹	3.4 ⁹⁸³	3.5 ⁷³⁹	3.4 ²⁰¹¹
Unsaturated HC	22	1.5 ⁹	0.8 ⁵³	1.9 ¹¹¹	1.5 ¹⁷³	22	7.1 ⁹	2.2 ⁵³	4.3 ¹¹¹	3.8 ¹⁷³
n-Alkanes	25	2.2 ²⁵¹	1.0 ⁵³⁶	1.2 ²⁷⁷	1.3 ¹⁰⁶⁴	25	2.9 ²⁵¹	1.9 ⁵³⁶	3.8 ²⁷⁷	2.6 ¹⁰⁶⁴
Alkanes (non-cyclic)	63	2.3 ²⁷⁶	1.0 ⁷¹⁴	1.7 ⁴³⁴	1.5 ¹⁴²⁴	63	2.8 ²⁷⁶	3.8 ⁷¹⁴	4.2 ⁴³⁴	3.7 ¹⁴²⁴
Alkanes (cyclic)	28	7.5 ⁴	1.0 ²¹⁶	0.9 ¹⁹⁴	1.0 ⁴¹⁴	28	0.4 ⁴	2.4 ²¹⁶	1.7 ¹⁹⁴	2.1 ⁴¹⁴
Aromatic HC	34	4.7 ³¹	1.3 ⁶⁷⁹	1.7 ¹³⁸	1.5 ⁸⁴⁸	34	20.5 ³¹	3.0 ⁶⁷⁹	2.8 ¹³⁸	3.6 ⁸⁴⁸
Fused aromatic HC	8	7.3 ²	1.7 ⁹³	1.8 ⁶⁴	1.8 ¹⁵⁹	8	11.3 ²	2.8 ⁹³	2.5 ⁶⁴	2.8 ¹⁵⁹
Alkenes HC	18	1.5 ⁹	0.8 ⁴⁵	2.0 ⁸¹	1.6 ¹³⁵	18	7.1 ⁹	2.5 ⁴⁵	5.3 ⁸¹	4.5 ¹³⁵
Alkenes (cyclic HC)	3	–	1.2 ⁷	1.9 ¹³	1.6 ²⁰	3	–	1.2 ⁷	7.7 ¹³	5.4 ²⁰
Alkynes HC	2	–	0.6 ⁷	4.0 ⁷	2.3 ¹⁴	2	–	0.4 ⁷	0.9 ⁷	0.7 ¹⁴

Table 13

Viscosity relative mean deviation (%) and reference temperature average mean deviation (K) of the proposed method for the different types of oxygen (alcohols, diols and triols) compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average; (AAD) average absolute deviation (K); (MAPE) mean average percentage error (%).

	NC	RMD (%) in $\ln(\eta)$				AMD in T_v		
		ELP	LP	MP	AV	NC	AAD	MAPE
1-Alcohols	16	4.8 ¹³³	6.4 ⁶¹⁰	2.1 ⁹⁴	5.7 ⁸³⁷	15	8.3	2.2
Alcohols ^a	123	7.7 ³¹⁷	6.6 ¹⁸⁴⁹	3.3 ⁴⁴⁰	6.2 ²⁶⁰⁶	119	10.8	3.0
Primary alcohols	29	5.1 ¹³⁶	6.7 ⁸²¹	3.2 ¹³⁶	6.1 ¹⁰⁹³	27	10.4	3.0
Secondary alcohols	29	21.2 ⁴	7.5 ²⁵⁹	1.8 ¹²⁹	5.8 ³⁹²	29	7.1	2.1
Tertiary alcohols	8	–	10.2 ⁵⁵	7.9 ⁶⁵	8.9 ¹²⁰	7	8.1	2.5
Aromatic alcohols	6	–	8.9 ¹²⁹	4.2 ¹⁰	8.5 ¹³⁹	6	7.2	1.9
Alkane diols, triols	13	10.7 ⁶¹	5.1 ²⁶⁴	0.0 ¹³	5.9 ³³⁸	12	9.2	2.3

^a Includes multi-functional compounds.

Table 14

Viscosity relative mean deviation (%) of the Andrade and Vogel model for the different types of oxygen compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average.

	NC	RMD (%) in $\ln(\eta)$									
		Andrade					Vogel				
		ELP	LP	MP	AV		NC	ELP	LP	MP	AV
1-Alcohols	16	2.4 ¹³³	1.8 ⁶¹⁰	2.0 ⁹⁴	1.9 ⁸³⁷	16	3.3 ¹³³	2.7 ⁶¹⁰	3.6 ⁹⁴	2.9 ⁸³⁷	
Alcohols ^a	123	3.9 ³¹⁷	2.8 ¹⁸⁴⁹	3.9 ⁴⁴⁰	3.2 ²⁶⁰⁶	111	4.4 ³⁰⁴	3.7 ¹⁷⁷⁷	5.8 ⁴²⁴	4.1 ²⁵⁰⁵	
Primary alcohols	29	2.5 ¹³⁶	2.1 ⁸²¹	2.3 ¹³⁶	2.2 ¹⁰⁹³	27	3.3 ¹³⁶	2.7 ⁸¹³	3.3 ¹³⁰	2.9 ¹⁰⁷⁹	
Secondary alcohols	29	4.3 ⁴	4.4 ²⁵⁹	3.8 ¹²⁹	4.2 ³⁹²	29	0.7 ⁴	3.3 ²⁵⁹	2.9 ¹²⁹	3.1 ³⁹²	
Tertiary alcohols	8	–	5.1 ⁵⁵	3.4 ⁶⁵	4.2 ¹²⁰	7	–	2.1 ⁵⁰	2.0 ⁶⁴	2.1 ¹¹⁴	
Aromatic alcohols	6	–	4.6 ¹²⁹	6.5 ¹⁰	4.8 ¹³⁹	6	–	3.0 ¹²⁹	2.8 ¹⁰	3.0 ¹³⁹	
Alkane diols, triols	13	4.1 ⁶¹	2.5 ²⁶⁴	14.4 ¹³	3.2 ³³⁸	11	10.0 ⁵⁵	9.4 ²⁴⁰	56.9 ¹¹	11.2 ³⁰⁶	

^a Includes multi-functional compounds.

For the viscosity reference temperature, the proposed method yielded a consistent set of results for the different classes of halogen compounds. For all halogenated compounds with only one halogen group, there were no deviations greater than 17 K. For polyhalogenated compounds, there were large deviations for halogenated silicon compounds. These components will be discussed in the next section. Overall, large deviations were only observed for diiodomethane (20 K) and tribromofluoromethane (26.8 K) and since these are the first in their respective series, they were removed from the regression set. Caution should always be taken when estimating the viscosity of highly halogenated methane and to some degree ethane compounds.

3.3.5. Various other compounds

Results for the various other types of compounds for the proposed method and both correlative models are presented in

Tables 23 and 24, respectively. There were also no especially large deviations from the estimation of the slope for these compounds over the entire temperature range.

Multiple plots of phosphate, boron, silicon and acid chloride compounds are presented in Figs. 3–6, respectively. The proposed method yielded an excellent agreement between estimated and experimental data and similar results are observed for the other compounds not plotted. For phosphate compounds, a slightly odd curvature of the trends from experimental data is observed for the latter three compounds in the plot. All data for these components were reported in the same publication. For boron compounds, a difference in the slope between the experimental and estimated trends is observed only for boric acid trimethyl ester which is also the smallest compound. However, estimations of the slope from other boron compounds show a better agreement. For silicon compounds, a slight disparity is observed for several compounds with

Table 15

Viscosity relative mean deviation (%) and reference temperature average mean deviation (K) of the proposed method for the different types of oxygen (except alcohol) compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average; (AAD) average absolute deviation (K); (MAPE) mean average percentage error (%).

	NC	RMD (%) in $\ln(\eta)$				AMD in T_v		
		ELP	LP	MP	AV	NC	AAD	MAPE
Ethers	27	8.8 ¹⁰	3.1 ²⁰⁸	2.8 ²¹⁵	3.1 ⁴³³	26	4.2	1.8
Epoxides	2	–	0.4 ⁵	0.7 ¹³	0.6 ¹⁸	2	4.3	2.3
Aldehydes	8	–	0.6 ²¹	0.8 ⁴⁴	0.8 ⁶⁵	7	5.2	2.3
Ketones	23	12.6 ¹³	2.5 ²³⁷	2.9 ¹⁸⁸	3.0 ⁴³⁸	23	7.9	2.8
Non-cyclic carbonates	3	1.8 ⁵⁶	0.9 ²⁴	1.5 ¹⁰	1.5 ⁹⁰	3	3.7	1.5
Carboxylic acids	22	4.6 ¹⁴⁸	1.9 ¹⁹⁴	3.8 ⁴⁴	3.1 ³⁸⁶	18	14.6	3.9
Esters	49	3.7 ⁴²	1.9 ⁴⁶⁸	1.5 ²⁷⁶	1.8 ⁷⁸⁶	49	5.2	1.9
Formic acids esters	9	–	2.3 ³¹	1.5 ⁹¹	1.7 ¹²²	9	3.4	1.4
Lactones	1	2.1 ⁸	1.2 ²⁰	0.0 ¹	1.4 ²⁹	1	0.0	0.0
Anhydride chains	4	–	0.9 ⁴⁷	0.5 ¹⁵	0.8 ⁶²	4	2.0	0.7
Anhydride cyclic	2	–	3.9 ⁵	0.0 ²	2.8 ⁷	2	12.2	3.2
Aromatic oxygen	1	–	–	0.0 ⁵	0.0 ⁵	1	0.0	0.0
All (w/o alcohols) ^a	254	5.4 ⁴⁷²	2.5 ¹⁸⁷⁸	2.2 ¹¹⁴⁵	2.8 ³⁴⁹⁵	248	7.8	2.5

^a Includes multi-functional compounds.

Table 16

Viscosity relative mean deviation (%) of the Andrade and Vogel model for the different types of oxygen (except alcohol) compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average.

	RMD (%) in $\ln(\eta)$									
	Andrade					Vogel				
	NC	ELP	LP	MP	AV	NC	ELP	LP	MP	AV
Ethers	27	1.8 ¹⁰	1.4 ²⁰⁸	1.2 ²¹⁵	1.3 ⁴³³	26	1.6 ¹⁰	1.6 ²⁰⁶	1.8 ²¹⁴	1.7 ⁴³⁰
Epoxides	2	–	0.4 ⁵	1.2 ¹³	0.9 ¹⁸	2	–	0.4 ⁵	5.3 ¹³	4.0 ¹⁸
Aldehydes	8	–	0.6 ²¹	1.7 ⁴⁴	1.4 ⁶⁵	7	–	0.6 ¹⁹	14.5 ⁴³	10.2 ⁶²
Ketones	23	2.3 ¹³	1.6 ²³⁷	1.5 ¹⁸⁸	1.5 ⁴³⁸	20	1.5 ¹³	1.7 ²²⁴	2.1 ¹⁸⁵	1.9 ⁴²²
Non-cyclic carbonates	3	2.0 ⁵⁶	0.7 ²⁴	2.2 ¹⁰	1.6 ⁹⁰	3	0.9 ⁵⁶	0.8 ²⁴	64.7 ¹⁰	7.9 ⁹⁰
Carboxylic acids	22	1.1 ¹⁴⁸	0.9 ¹⁹⁴	5.8 ⁴⁴	1.6 ³⁸⁶	21	1.4 ¹⁴⁸	1.5 ¹⁹²	5.5 ⁴³	1.9 ³⁸³
Esters	49	2.1 ⁴²	1.0 ⁴⁶⁸	1.3 ²⁷⁶	1.2 ⁷⁸⁶	42	1.5 ⁴¹	1.5 ⁴⁵⁵	2.5 ²⁶¹	1.9 ⁷⁵⁷
Formic acids esters	9	–	1.0 ³¹	1.2 ⁹¹	1.1 ¹²²	9	–	4.0 ³¹	5.9 ⁹¹	5.4 ¹²²
Lactones	1	2.4 ⁸	1.2 ²⁰	0.0 ¹	1.5 ²⁹	1	1.8 ⁸	0.8 ²⁰	3.7 ¹	1.2 ²⁹
Anhydride chains	4	–	1.0 ⁴⁷	0.4 ¹⁵	0.8 ⁶²	3	–	0.9 ⁴⁵	0.5 ¹⁴	0.9 ⁵⁹
Anhydride cyclic	2	–	0.2 ⁵	12.6 ²	3.7 ⁷	2	–	0.4 ⁵	11.9 ²	3.7 ⁷
Aromatic oxygen	1	–	–	0.4 ⁵	0.4 ⁵	1	–	–	0.4 ⁵	0.4 ⁵
All (w/o alcohols) ^a	254	2.6 ⁴⁷²	1.3 ¹⁸⁷⁸	2.0 ¹¹⁴⁵	1.7 ³⁴⁹⁵	207	2.0 ⁴⁵⁴	1.7 ¹⁷⁴⁵	3.9 ¹⁰⁶³	2.4 ³²⁶²

^a Includes multi-functional compounds.

Table 17

Viscosity relative mean deviation (%) and reference temperature average mean deviation (K) of the proposed method for the different types of nitrogen compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average; (AAD) average absolute deviation (K); (MAPE) mean average percentage error (%).

	NC	RMD (%) in $\ln(\eta)$				AMD in T_v		
		ELP	LP	MP	AV	NC	AAD	MAPE
Primary amines	22	16.8 ²	4.0 ²⁷⁷	2.6 ¹⁵²	3.5 ⁴³¹	21	5.1	1.8
Secondary amines	15	8.8 ¹⁰	3.6 ¹³⁷	3.1 ⁵¹	3.7 ¹⁹⁸	15	2.2	0.7
Tertiary amines	6	8.6 ⁴	2.3 ¹²⁴	2.9 ⁴³	2.6 ¹⁷¹	6	7.0	2.4
Amines ^a	59	11.6 ⁶⁵	4.6 ⁵⁷⁰	3.0 ²⁷¹	4.6 ⁹⁰⁶	58	5.9	1.8
N in 5-membered rings	2	–	2.3 ³¹	0.0 ²	2.1 ³³	2	1.2	0.4
N in 6-membered rings	16	6.3 ¹⁶	3.4 ²¹³	3.9 ³²	3.6 ²⁶¹	16	2.7	0.9
Cyanides	15	4.2 ⁷	2.1 ¹⁴²	1.7 ¹⁷⁴	2.0 ³²³	14	6.5	2.5
Amides	3	–	2.8 ⁵⁹	0.4 ⁴	2.6 ⁶³	2	1.2	0.3
Mono amides	3	6.4 ¹¹	6.6 ⁵⁹	0.0 ³	6.3 ⁷³	2	8.8	2.4
Di amides	5	6.8 ⁶	1.3 ¹⁶⁰	0.0 ⁵	1.5 ¹⁷¹	5	8.0	2.8
Isocyanates	4	–	4.5 ¹⁵	2.7 ¹⁶	3.6 ³¹	4	9.6	4.2
Oximes	1	–	0.0 ²	0.0 ¹	0.0 ³	1	–	–
Nitrous and nitrites	14	4.0 ¹²	2.4 ¹⁵⁸	4.0 ⁷³	3.0 ²⁴³	14	4.7	1.7
Nitrates	1	–	0.0 ²	0.0 ¹	0.0 ³	1	–	–
All (w/o amines)	62	5.5 ⁵¹	2.7 ⁷⁴⁵	1.9 ²⁵⁴	2.7 ¹⁰⁵⁰	59	5.0	1.8

^a Includes multi-functional compounds.

Table 18

Viscosity relative mean deviation (%) of the Andrade and Vogel model for the different types of nitrogen compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average.

	RMD (%) in $\ln(\eta)$									
	Andrade					Vogel				
	NC	ELP	LP	MP	AV	NC	ELP	LP	MP	AV
Primary amines	22	13.2 ²	2.1 ²⁷⁷	1.8 ¹⁵²	2.1 ⁴³¹	20	2.3 ²	1.4 ²⁷⁵	6.6 ¹⁴⁴	3.2 ⁴²¹
Secondary amines	15	3.4 ¹⁰	2.7 ¹³⁷	3.9 ⁵¹	3.0 ¹⁹⁸	13	2.3 ⁹	2.1 ¹³³	5.0 ⁴⁹	2.8 ¹⁹¹
Tertiary amines	6	4.2 ⁴	1.8 ¹²⁴	1.3 ⁴³	1.7 ¹⁷¹	6	3.2 ⁴	1.7 ¹²⁴	1.6 ⁴³	1.7 ¹⁷¹
Amines ^a	59	4.8 ⁶⁵	2.6 ⁵⁷⁰	2.6 ²⁷¹	2.7 ⁹⁰⁶	49	2.9 ⁶⁰	1.8 ⁵²⁹	7.0 ²⁴⁶	3.4 ⁸³⁵
N in 5-membered rings	2	–	2.3 ³¹	1.5 ²	2.3 ³³	2	–	2.3 ³¹	1.9 ²	2.3 ³³
N in 6-membered rings	16	3.1 ¹⁶	1.1 ²¹³	2.3 ³²	1.4 ²⁶¹	15	2.8 ¹⁶	1.5 ²¹¹	3.3 ³¹	1.8 ²⁵⁸
Cyanides	15	3.4 ⁷	1.2 ¹⁴²	1.8 ¹⁷⁴	1.6 ³²³	11	3.6 ⁴	1.1 ¹³⁷	19.7 ¹⁷⁰	11.3 ³¹¹
Amides	3	–	1.6 ⁵⁹	4.0 ⁴	1.7 ⁶³	3	–	2.5 ⁵⁹	3.8 ⁴	2.6 ⁶³
Mono amides	3	1.0 ¹¹	1.7 ⁵⁹	14.3 ³	2.1 ⁷³	3	1.2 ¹¹	1.6 ⁵⁹	4.9 ³	1.7 ⁷³
Di amides	5	0.6 ⁶	0.9 ¹⁶⁰	2.0 ⁵	0.9 ¹⁷¹	4	0.5 ⁵	0.9 ¹⁵⁷	0.7 ⁴	0.9 ¹⁶⁶
Isocyanates	4	–	0.4 ¹⁵	5.2 ¹⁶	2.9 ³¹	2	–	0.3 ⁸	2.9 ⁷	1.5 ¹⁵
Oximes	1	–	0.0 ²	3.3 ¹	1.1 ³	–	–	–	–	–
Nitrous and nitrites	14	1.2 ¹²	1.1 ¹⁵⁸	1.2 ⁷³	1.1 ²⁴³	14	0.8 ¹²	1.5 ¹⁵⁸	2.1 ⁷³	1.6 ²⁴³
Nitrates	1	–	0.0 ²	1.1 ¹	0.4 ³	–	–	–	–	–
All (w/o amines)	62	1.9 ⁵¹	1.2 ⁷⁴⁵	2.4 ²⁵⁴	1.5 ¹⁰⁵⁰	50	1.9 ⁴⁷	1.4 ⁷¹⁴	15.2 ²³⁵	4.7 ⁹⁹⁶

^a Includes multi-functional compounds.

Table 19

Viscosity relative mean deviation (%) and reference temperature average mean deviation (K) of the proposed method for the different types of sulphur compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average; (AAD) average absolute deviation (K); (MAPE) mean average percentage error (%).

	NC	RMD (%) in $\ln(\eta)$				AMD in T_v		
		ELP	LP	MP	AV	NC	AAD	MAPE
Disulfides	2	–	0.1 ⁶	0.0 ²	0.1 ⁸	2	2.6	1.0
Thiols	12	–	1.5 ³⁹	1.7 ³⁶	1.6 ⁷⁵	12	7.2	2.9
Thioether	8	–	0.9 ²⁹	0.6 ²⁷	0.8 ⁵⁶	8	6.2	2.6
Aromatic thioether	3	–	0.8 ¹⁷	1.2 ¹⁹	1.0 ³⁶	3	1.5	0.6
Sulfolane (O=S=O)	3	1.3 ⁵¹	4.9 ¹⁰	0.0 ³	1.8 ⁶⁴	3	6.6	1.6
Isothiocyanates	2	–	1.6 ⁹	0.0 ²	1.3 ¹¹	2	0.9	0.4
Sulfates, sulfon amides and sulfoxides	6	4.1 ¹⁵	1.8 ¹⁰	0.0 ⁶	2.6 ³¹	6	3.8	1.1
Sulphur compounds ^a	41	2.0 ⁶⁶	1.4 ¹³⁸	1.0 ¹⁰²	1.4 ³⁰⁶	41	6.1	2.2

^a Includes multi-functional compounds.

Table 20

Viscosity relative mean deviation (%) of the Andrade and Vogel model for the different types of sulphur compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average.

	RMD (%) in $\ln(\eta)$									
	Andrade					Vogel				
	NC	ELP	LP	MP	AV	NC	ELP	LP	MP	AV
Disulfides	2	–	0.0 ⁶	1.1 ²	0.3 ⁸	2	–	0.0 ⁶	19.2 ²	4.8 ⁸
Thiols	12	–	0.6 ³⁹	2.3 ³⁶	1.4 ⁷⁵	12	–	2.0 ³⁹	7.2 ³⁶	4.5 ⁷⁵
Thioether	8	–	0.5 ²⁹	1.8 ²⁷	1.1 ⁵⁶	8	–	0.5 ²⁹	5.1 ²⁷	2.7 ⁵⁶
Aromatic thioether	3	–	0.6 ¹⁷	0.9 ¹⁹	0.8 ³⁶	3	–	0.6 ¹⁷	0.8 ¹⁹	0.7 ³⁶
Sulfolane (O=S=O)	3	0.7 ⁵¹	1.6 ¹⁰	10.6 ³	1.3 ⁶⁴	3	1.0 ⁵¹	0.7 ¹⁰	20.6 ³	1.8 ⁶⁴
Isothiocyanates	2	–	0.8 ⁹	6.0 ²	1.8 ¹¹	2	–	0.8 ⁹	15.1 ²	3.4 ¹¹
Sulfates, sulfon amides and sulfoxides	6	2.1 ¹⁵	1.8 ¹⁰	5.9 ⁶	2.7 ³¹	1	4.2 ³	0.7 ⁵	0.9 ¹	1.9 ⁹
Sulphur compounds ^a	41	1.0 ⁶⁶	0.7 ¹³⁸	2.4 ¹⁰²	1.3 ³⁰⁶	35	1.1 ⁵⁴	1 ¹³¹	5.9 ⁹⁶	2.7 ²⁸¹

^a Includes multi-functional compounds.

Table 21

Viscosity relative mean deviation (%) and reference temperature average mean deviation (K) of the proposed method for the different types of halogen compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average; (AAD) average absolute deviation (K); (MAPE) mean average percentage error (%).

	NC	RMD (%) in $\ln(\eta)$				AMD in T_v		
		ELP	LP	MP	AV	NC	AAD	MAPE
Saturated fluorine	13	–	3.3 ²⁸	4.2 ¹⁵¹	4.1 ¹⁷⁹	13	6.1	3.1
Fluorinated	20	–	2.3 ⁵⁴	3.5 ¹⁹⁸	3.3 ²⁵²	20	8.0	3.7
Saturated chlorine	34	4.5 ⁶	1.6 ¹⁶¹	2.6 ⁴³⁹	2.4 ⁶⁰⁶	34	7.3	2.9
Chlorinated	50	2.3 ¹⁵	1.4 ³⁵⁷	2.6 ⁵⁴⁰	2.1 ⁹¹²	50	7.4	2.9
Saturated bromine	21	–	1.6 ²³¹	1.7 ¹⁴³	1.6 ³⁷⁴	21	6.6	2.2
Brominated	28	4.3 ³	1.7 ³¹⁷	1.8 ¹⁵⁵	1.8 ⁴⁷⁵	28	6.4	2.1
Iodinated	14	9.5 ¹³	2.2 ⁷⁹	2.3 ⁶⁰	2.9 ¹⁵²	13	3.8	1.4
Halogenated compounds ^a	182	7.8 ⁶⁹	2.3 ¹²¹⁰	2.6 ¹³⁸⁵	2.6 ²⁶⁶⁴	179	7.1	2.8

^a Includes multi-functional compounds.

Table 22

Viscosity relative mean deviation (%) of the Andrade and Vogel model for the different types of halogen compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average.

	RMD (%) in $\ln(\eta)$									
	Andrade					Vogel				
	NC	ELP	LP	MP	AV	NC	ELP	LP	MP	AV
Saturated fluorine	13	–	1.4 ²⁸	1.6 ¹⁵¹	1.5 ¹⁷⁹	12	–	1.5 ²⁸	2.5 ¹⁴⁶	2.4 ¹⁷⁴
Fluorinated	20	–	1.0 ⁵⁴	1.6 ¹⁹⁸	1.5 ²⁵²	19	–	1.0 ⁵⁴	2.3 ¹⁹³	2.0 ²⁴⁷
Saturated chlorine	34	0.9 ⁶	0.9 ¹⁶¹	0.9 ⁴³⁹	0.9 ⁶⁰⁶	29	0.3 ⁶	1.7 ¹⁵⁵	2.0 ⁴²³	1.9 ⁵⁸⁴
Chlorinated	50	0.6 ¹⁵	0.8 ³⁵⁷	0.9 ⁵⁴⁰	0.8 ⁹¹²	45	1.8 ¹⁵	1.2 ³⁵¹	1.9 ⁵²⁴	1.6 ⁸⁹⁰
Saturated bromine	21	–	0.9 ²³¹	0.9 ¹⁴³	0.9 ³⁷⁴	20	–	1.1 ²²⁹	1.7 ¹⁴²	1.3 ³⁷¹
Brominated	28	0.5 ³	0.9 ³¹⁷	1.2 ¹⁵⁵	1.0 ⁴⁷⁵	26	26.2 ¹	1.3 ³¹⁵	2.3 ¹⁵³	1.7 ⁴⁶⁹
Iodinated	14	3.6 ¹³	2.1 ⁷⁹	1.3 ⁶⁰	1.9 ¹⁵²	12	4.0 ¹²	2.4 ⁷⁶	1.9 ⁵⁸	2.4 ¹⁴⁶
Halogenated compounds ^a	182	2.5 ⁶⁹	1.3 ¹²¹⁰	1.1 ¹³⁸⁵	1.3 ²⁶⁶⁴	154	4.1 ⁶²	1.6 ¹¹³⁵	2.1 ¹³⁰⁹	1.9 ²⁵⁰⁶

^a Includes multi-functional compounds.

Table 23

Viscosity relative mean deviation (%) and reference temperature average mean deviation (K) of the proposed method for the various other types of compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average; (AAD) average absolute deviation (K); (MAPE) mean average percentage error (%).

	NC	RMD (%) in $\ln(\eta)$				AMD in T_v		
		ELP	LP	MP	AV	NC	AAD	MAPE
Phosphates	4	3.8 ¹⁰⁶	5.2 ²	0.0 ⁴	3.7 ¹¹²	4	17.0	4.6
Germanium	1	–	–	0.1 ⁷	0.1 ⁷	1	0.0	0.0
Boron	6	0.5 ²	1.3 ³⁰	2.0 ²⁹	1.6 ⁶¹	6	9.8	4.2
Silicon	2	–	0.5 ⁵	0.0 ²	0.4 ⁷	2	9.8	3.9
Silicon (<i>en</i> ^a)	23	4.5 ¹⁸	2.8 ⁷⁹	2.2 ⁷³	2.7 ¹⁷⁰	22	17.7	6.2
Acid chloride	5	–	1.6 ⁷	1.1 ²⁴	1.2 ³¹	5	1.8	0.8
Urea	1	–	0.3 ⁹	0.0 ¹	0.3 ¹⁰	1	0.0	0.0

^a Denotes silicon connected to any electronegative neighbor. This filter also includes all multi-functional compounds.

Table 24

Viscosity relative mean deviation (%) of the Andrade and Vogel model for various other types of compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average.

	RMD (%) in $\ln(\eta)$									
	Andrade					Vogel				
	NC	ELP	LP	MP	AV	NC	ELP	LP	MP	AV
Phosphates	4	2.1 ¹⁰⁶	4.8 ²	10.9 ⁴	2.5 ¹¹²	4	5.2 ¹⁰⁶	3.3 ²	17.3 ⁴	5.6 ¹¹²
Germanium	1	–	–	0.3 ⁷	0.3 ⁷	1	–	–	0.8 ⁷	0.8 ⁷
Boron	6	1.2 ²	0.8 ³⁰	2.5 ²⁹	1.6 ⁶¹	6	0.3 ²	0.6 ³⁰	3.4 ²⁹	1.9 ⁶¹
Silicon	2	–	0.0 ⁵	0.9 ²	0.3 ⁷	–	–	–	–	–
Silicon (<i>en</i> ^a)	23	1.1 ¹⁸	1.3 ⁷⁹	2.6 ⁷³	1.9 ¹⁷⁰	19	0.8 ¹⁵	1.2 ⁷⁶	8.0 ⁶⁷	4.0 ¹⁵⁸
Acid chloride	5	–	0.2 ⁷	1.7 ²⁴	1.4 ³¹	4	–	0.3 ⁵	2.7 ²³	2.3 ²⁸
Urea	1	–	0.2 ⁹	0.0 ¹	0.2 ¹⁰	1	–	0.8 ⁹	1.0 ¹	0.8 ¹⁰

^a Denotes silicon connected to any electronegative neighbor. This filter also includes all multi-functional compounds.

data from only one source of data, such as trimethylchlorosilane and hexamethyldisiloxane.

For the viscosity reference temperature, a large error was only observed for triphenyl phosphate (30.8 K). Phosphate triester is the only phosphate compound in the training set where the oxygen atoms are connected to an aromatic carbon. Usually, a distinction is required here; however, this distinction was not observed for the estimation of the normal boiling point of these compounds (an estimation error of 4 K was reported). Considering the questionable nature of the data and since there is only one component, a new group was not added in this case.

There were also large deviations observed for silicon compounds, especially for smaller highly halogenated or oxygenated compounds. The largest deviations reported were for trichlorophenylsilane (45.0 K), octadecamethyloctasiloxane (35.0 K) and trimethylchlorosilane (30.0 K). For these compounds,

the greater steric strain and subsequent change in polarizability, especially in the case of smaller compounds leads to larger deviations. With increasing molecular weight, the estimation improves and extrapolates correctly as shown in Fig. 7. Trimethylchlorosilane was also the first compound in its series and was removed from the regression set.

3.4. Multi-functional compounds

The estimation of the viscosity reference temperature for multi-functional compounds employs interaction groups as in the case of other properties. To account for the non-additivity of group contributions in case of molecules with more than one hydrogen bonding group, a group interaction contribution is added to the sum of group

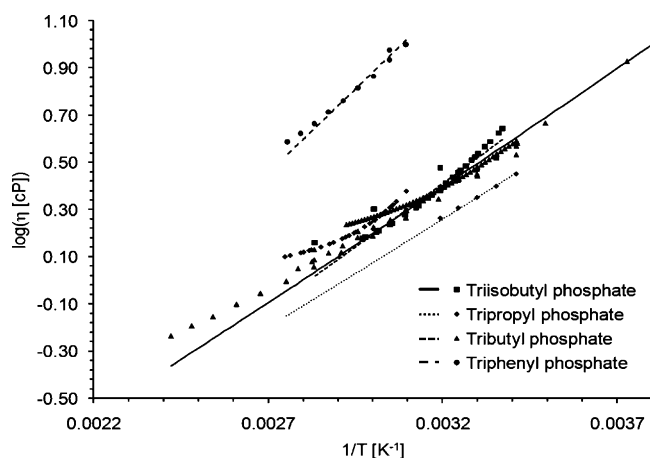


Fig. 3. Liquid viscosity estimation results for substituted phosphates together with experimental data from the DDB using adjusted reference temperatures.

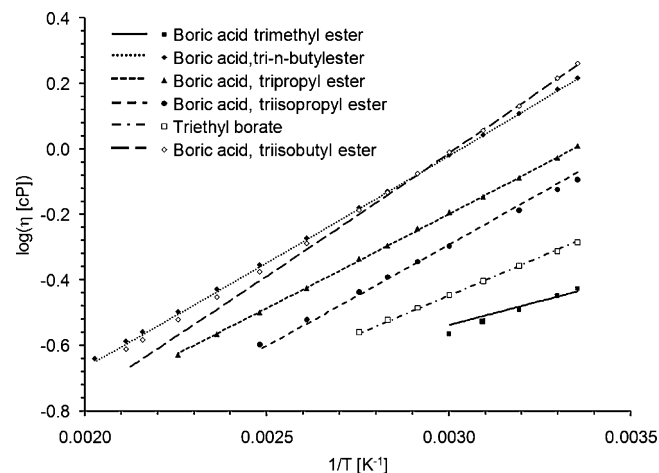


Fig. 4. Liquid viscosity estimation results for boric acid esters together with experimental data from the DDB using adjusted reference temperatures.

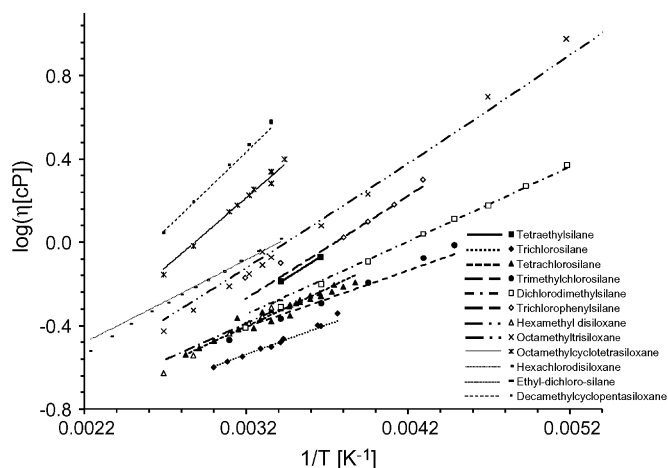


Fig. 5. Liquid viscosity estimation results for silanes and siloxanes together with experimental data from the DDB using adjusted reference temperatures.

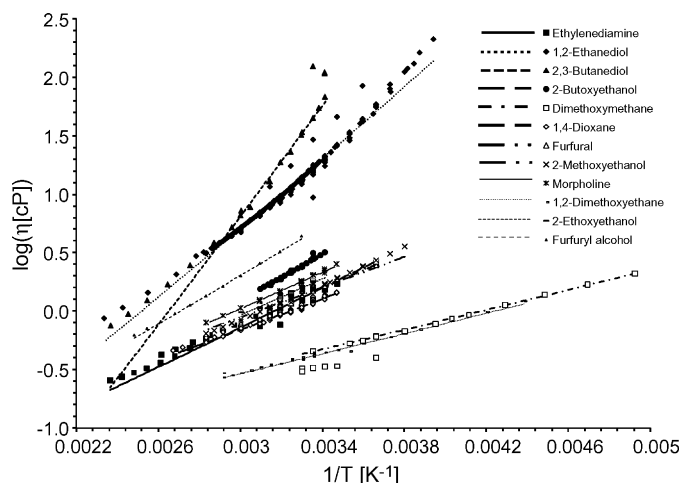


Fig. 8. Liquid viscosity estimation results for multifunctional components together with experimental data from the DDB using adjusted reference temperatures.

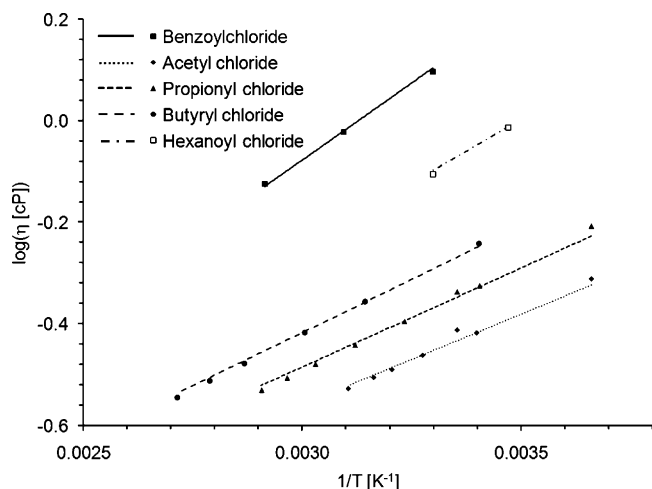


Fig. 6. Liquid viscosity estimation results for carbonyl chlorides together with experimental data from the DDB using adjusted reference temperatures.

contributions:

$$GI = \frac{f_{GI}}{n} \sum_{i=1}^m \sum_{j=1}^m \left(\frac{C_{i-j}}{m-1} \right) \quad \text{where } C_{i-j} = C_{j-i} \quad (10)$$

As described earlier, C_{i-j} is the group interaction contribution between group i and group j (where $C_{i-i} = 0$), n is the number of

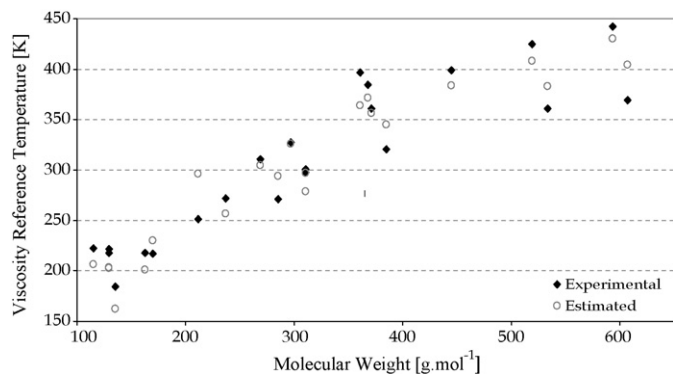


Fig. 7. Plot of viscosity reference temperatures for silicon compounds connected to electronegative atoms. "Experimental" denotes values derived from experimental data by inter- or extrapolation.

atoms (except hydrogen) and m is the total number of interaction groups in the molecule. The factor f_{GI} is equal to 2 in case of T_v estimation and 1 in case of dB_v estimation.

Results for multi-functional compounds for the proposed method and both correlative models are presented in Tables 25 and 26, respectively. Slightly larger average errors should be expected as the experimental information is often of low quality for these molecules. However, there were no cases of extreme deviations for all classes of compounds over the whole temperature ranges.

Multiple plots for multi-functional compounds are presented in Figs. 8–11. The proposed method yielded an excellent agreement between estimated and experimental data and similar results are observed for the other compounds not plotted.

The lower quality of the data in many cases leads to a scatter of the data points. In the case of ethylenediamine, Fig. 8, two different sources of data (Friend and Hargreaves [20] and Kapadi et al. [21]) show two different temperature trends. The former reference covers the higher temperature range and was found to be questionable for other compounds, e.g. alpha-aminotoluene. The latter reference is a recent measurement and shows a good agreement with the proposed method. The error between the unreliable reference and the proposed method increased the average deviation while correlative models produce a more accurate fit. There are also components where dissimilar viscosity values are reported for the same temper-

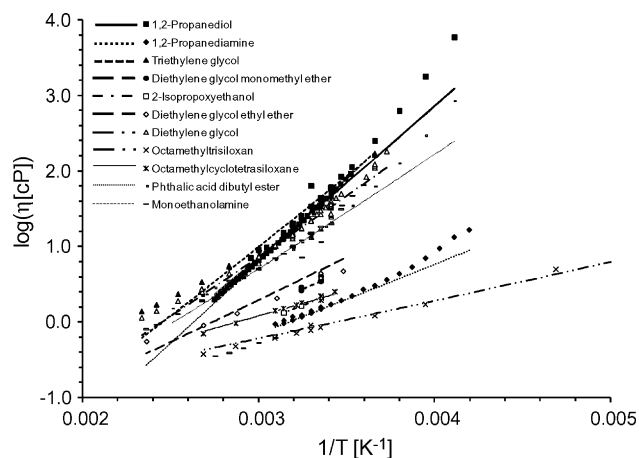


Fig. 9. Liquid viscosity estimation results for multifunctional components together with experimental data from the DDB using adjusted reference temperatures.

Table 25

Viscosity relative mean deviation (%) and reference temperature average mean deviation (K) of the proposed method for multi-functional compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average; (AAD) average absolute deviation (K); (MAPE) mean average percentage error (%).

	RMD (%) in $\ln(\eta)$					AMD in T_r		
	NC	ELP	LP	MP	AV	NC	AAD	MAPE
OH	36	9.4 ¹⁷¹	5.2 ⁴⁷⁸	0.8 ³⁷	6.0 ⁶⁸⁶	35	13.3	3.6
OH(a)	8	10.6 ⁶	4.9 ⁶⁶	3.8 ³⁰	4.9 ¹⁰²	8	20.9	5.4
NH ₂	10	15.6 ²	9.3 ⁷⁸	2.0 ¹²	8.5 ⁹²	10	7.0	1.9
NH	3	7.3 ¹⁸	3.3 ²⁵	0.0 ³	4.6 ⁴⁶	3	6.5	1.6
Ether	64	7.0 ⁹³	3.2 ⁴⁹⁴	2.9 ¹⁴⁹	3.6 ⁷³⁶	64	10.8	3.3
Ester	16	6.8 ¹⁰¹	3.5 ⁴⁰	0.0 ¹⁶	5.3 ¹⁵⁷	16	7.2	2.0
Ketone	5	–	0.3 ¹⁴	1.1 ⁷	0.6 ²¹	5	1.5	0.4
Aldehyde	2	3.6 ¹	0.3 ³	0.0 ²	0.7 ⁶	2	0.0	0.0
Nitro, CN, AO, AN6	11	2.1 ⁸	3.2 ⁸⁵	1.4 ¹⁵	2.9 ¹⁰⁸	11	9.3	2.4
All GI components ^a	111	8.1 ³¹²	4.1 ⁹⁴⁶	2.3 ²⁰⁵	4.7 ¹⁴⁶³	110	10.3	2.9

^a GI components are components where group interactions between non-additive groups occur.

Table 26

Viscosity relative mean deviation (%) of the Andrade and Vogel model for multi-functional compounds (number of data points as superscript): (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average.

	RMD (%) in $\ln(\eta)$									
	Andrade					Vogel				
	NC	ELP	LP	MP	AV	NC	ELP	LP	MP	AV
OH	36	4.8 ¹⁷¹	2.4 ⁴⁷⁸	11.9 ³⁷	3.5 ⁶⁸⁶	27	5.6 ¹⁵⁸	6.7 ⁴¹⁹	49.7 ²⁸	8.4 ⁶⁰⁵
OH(a)	8	11.3 ⁶	3.3 ⁶⁶	3.3 ³⁰	3.8 ¹⁰²	8	1.0 ⁶	1.8 ⁶⁶	2.7 ³⁰	2.0 ¹⁰²
NH ₂	10	7.9 ²	5.1 ⁷⁸	10.8 ¹²	5.9 ⁹²	8	0.4 ¹	3.4 ⁷²	9.7 ¹⁰	4.1 ⁸³
NH	3	4.4 ¹⁸	2.3 ²⁵	8.6 ³	3.5 ⁴⁶	2	3.0 ¹⁸	1.6 ¹⁶	60.6 ²	5.6 ³⁶
Ether	64	5.1 ⁹³	1.8 ⁴⁹⁴	2.9 ¹⁴⁹	2.4 ⁷³⁶	41	3.9 ⁷⁹	2.0 ⁴¹⁹	6.8 ¹⁰¹	3.1 ⁵⁹⁹
Ester	16	3.1 ¹⁰¹	0.8 ⁴⁰	13.2 ¹⁶	3.6 ¹⁵⁷	13	2.1 ⁹⁸	1.4 ³⁵	21.4 ¹³	3.7 ¹⁴⁶
Ketone	5	–	0.3 ¹⁴	1.4 ⁷	0.7 ²¹	1	–	0.2 ⁶	0.0 ¹	0.2 ⁷
Aldehyde	2	0.0 ¹	0.0 ³	0.1 ²	0.6 ⁶	1	–	3.5 ²	9.1 ¹	5.4 ³
Nitro, CN, AO, AN6	11	2.9 ⁸	1.7 ⁸⁵	2.2 ¹⁵	1.9 ¹⁰⁸	10	1.9 ⁸	1.8 ⁷⁶	5.0 ¹⁴	2.2 ⁹⁸
All GI components	111	4.1 ³¹²	2.1 ⁹⁴⁶	5.0 ²⁰⁵	2.9 ¹⁴⁶³	80	4.2 ²⁸⁹	4.2 ⁸²²	14.2 ¹⁴⁹	5.4 ¹²⁶⁰

ature. For example, for 1,4-dioxane (Fig. 8), four different viscosity values at the same temperature from four different references are shown. Unfortunately, none of these data points could be verified. Overall, the estimation is based on chemically similar compounds and the method can to a certain extent be employed to verify data.

Compounds with amine interaction groups usually showed the largest disparity and a higher deviation in Table 25. Most of the data were reported by Friend and Hargreaves who published questionable values in several other cases. Overall, the proposed method reports satisfactory results, even for components where the viscosity reference temperature was extrapolated.

3.5. Overall results

For the group contribution estimation of dB_V , the proposed method reported an average absolute deviation of 0.2 in $\ln(\eta)$ (3.3%) for 829 components. This estimation error is acceptable as there are a number of cases where the data are of poor quality. There were also no exceptionally high deviations (>26%).

The results for the estimation of liquid viscosities for all data points from this work and correlative models are presented in Table 27. Overall, the proposed method yields results that are in comparable accuracy to the correlative models.

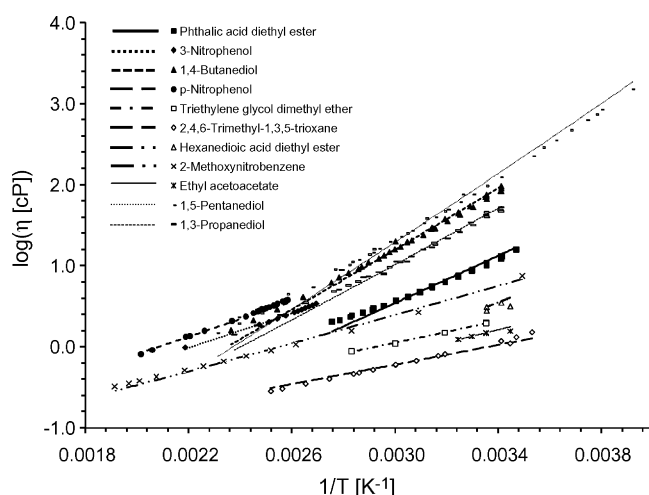


Fig. 10. Liquid viscosity estimation results for multifunctional components together with experimental data from the DDB using adjusted reference temperatures.

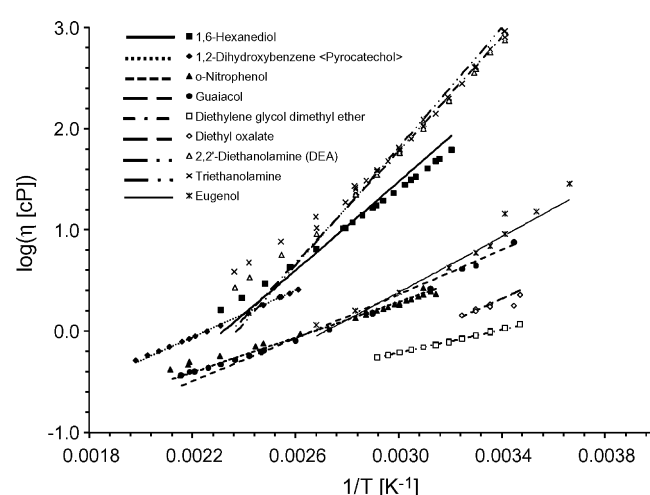


Fig. 11. Liquid viscosity estimation results for multifunctional components together with experimental data from the DDB using adjusted reference temperatures.

Table 27

Viscosity relative mean deviation (%) of the proposed method and correlative models for all compounds: (NC) number of components; ELP (extremely low pressure) $P < 0.01$ kPa; LP (low pressure) $0.01 \text{ kPa} \leq P \leq 10 \text{ kPa}$; MP (medium to higher pressure) $P > 10 \text{ kPa}$; (AV) average.

	RMD (%) in $\ln(\eta)$				
	NC	ELP	LP	MP	AV
This work	829	5.6 ¹³⁶³	3.5 ⁷⁸⁹⁶	2.5 ⁴⁴³¹	3.4 ¹³⁶⁹⁰
Andrade	829	2.6 ¹³⁶³	1.6 ⁷⁸⁹⁶	1.8 ⁴⁴³¹	1.8 ¹³⁶⁹⁰
Vogel	723	3.0 ¹³⁰⁴	2.1 ⁷⁶⁰⁶	4.3 ⁴²⁴⁴	2.9 ¹³¹⁵⁴

The proposed method may yield a slightly higher deviation as compared to the correlative models, but some errors are attributed to the inconsistent and unreliable experimental data. These errors are usually for components where the data were taken from older references. For many components, data from only one reference were available.

4. Data base preparation

Liquid viscosity data were taken from the Dortmund Data Bank (DDB) [6]. The DDB contains approximately 103,000 viscosity data points from 2630 references and approximately 2400 components, but not all of these data are at or near saturation pressure. For many components only one data point is available which does not allow regression of both the slope and reference temperature. During this work, the available data were carefully checked and numerous questionable data were not considered in the model development. Altogether saturated liquid viscosity data for approximately 830 components were used.

5. Test of the predictive capability

As there were significantly fewer data points available for liquid viscosity as compared to liquid vapour pressure, all data points were used in the regression and no separate test set was prepared. During the development of the liquid viscosity method, no significant qualitative differences to the modelling of the liquid vapour pressure and critical property data were observed. It is therefore considered improbable that the method for liquid viscosity estimation would perform significantly different when applied to a test set. In order to be able to test the method on components outside the training set, a large number of estimations of liquid viscosity at the normal boiling temperature were tested against an empirical rule (described in the sections that follow).

6. Liquid viscosity at the normal boiling temperature

It has been observed by many researchers in the past, that the liquid viscosity at the normal boiling temperature of a component usually falls into a rather narrow range. A recent overview with updated recommendations was given by Smith et al. [7]. In order to further verify the validity of the estimation method proposed in this work, liquid viscosity was estimated for a set of 4192 components, for which mostly reliable boiling point information was available. Fig. 12 shows the distribution of liquid viscosity values.

While more than 87% of the components obey the empirical rule of Smith et al., 2.6% show a higher and 10% a lower viscosity. It should be noted that for this calculation both the slope and the viscosity reference temperature were estimated. The uncertainty in the estimation of the reference temperature especially leads to a broadening of the distribution in Fig. 12. The empirical criterion of Smith et al. should be used when estimating the viscosity of unknown components. Modifying the reference temperature

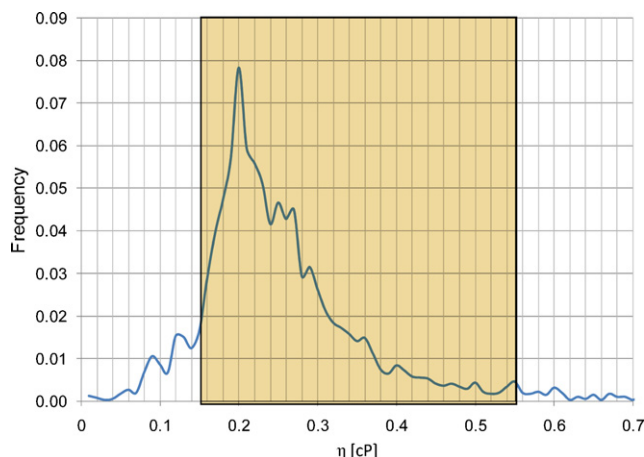


Fig. 12. Frequency of liquid viscosity values at the normal boiling point for a set of 4192 components (data taken from the DDB [6]).

within its probable estimation error in order to satisfy the criterion may well improve the estimation.

7. Reference temperature omission

Using a single experimental liquid viscosity allows one to calculate the viscosity reference temperature from Eq. (6) using an estimated value of dBv . For the components in the training set, back calculation of the viscosity reference temperature produced an average absolute deviation of 3.1 K (ELP), 1.6 K (LP), 3.0 K (MP) and 1.8 K for all points, respectively.

If there are no viscosity data available, two options can be used to estimate the reference temperature. The first is the empirical method of Smith et al. [7]. An alternate method is to employ the group contribution method proposed in this paper. For this method, an average absolute deviation of 7.1 K (2.5%) was obtained for 813 components. As expected, this error is in the same order of magnitude of the error in normal boiling temperature estimation observed in our previous work [2].

If the estimated viscosity reference temperature is used instead of the adjusted value to estimate the liquid viscosity using the method proposed in this work, a relative absolute deviation of 15.3% in viscosity is obtained for the 813 components or 12,139 data points in the training set. For the Van Velzen method, an average absolute deviation of 92.8% was obtained for 670 components or 11,115 data points. The method of Van Velzen leads to extremely large deviations for compounds that were probably not in the training set. In the case of the other group contribution methods (see Table 1), much higher errors and in some cases disastrous estimations were found. The method of Sastri and Rao for example produced an error of greater than 800% for tertiary and multi-functional alcohols and even larger errors for fluorine compounds.

8. Probability of prediction failure

The probability of predicting the viscosity reference temperature within a given error is plotted in Fig. 13. Ten percent of the estimations show a deviation greater than approximately 16 K. The larger errors were almost solely for smaller compounds or compounds that are the first members in their respective homologous series. First members generally do not follow the trend of the series. Data are usually available for these components and even if there is only a single point, back calculation of the viscosity reference temperature using an estimated slope is mostly more reliable than estimation.

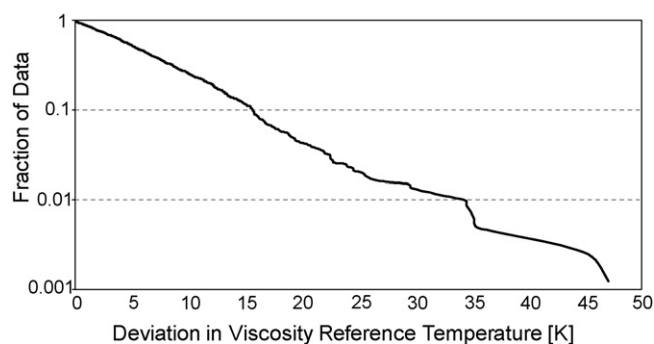


Fig. 13. Fraction of the data with deviations for viscosity reference temperature larger than a given value.

Estimations in case of multi-functional compounds that are strongly influenced by mesomeric and inductive effects (e.g. substituted nitroalkanes) were generally found to be less reliable.

In the case of slope (dBv) estimation, slightly higher errors were only observed for small molecules in the amine, nitro and alcohol series. Data for these components were often from less reliable sources and errors may easily result from the limited quality of the measurements.

9. Conclusion

A new group contribution method has been developed for the prediction of saturated liquid viscosity data. The method requires knowledge of only the molecular structure. Results are usually improved if one reliable experimental data point is available. Overall, the proposed method yielded results that are in comparable accuracy to equations regressed to experimental data and shows an excellent agreement between estimated and experimental data with no significant probability of prediction failure observed.

The core parameters for frequent groups like CH, CH₂, etc., can be considered final and missing or questionable contributions can be regressed as new data become available.

List of Symbols

A, B, C, a, b, c, d, e, s adjustable parameters

C_{i-j} group interaction contribution between group i and group j

$C(..)$ group contribution

dBv slope parameter

GI total group interaction contribution

$\ln$ natural logarithm

m number of interacting (hydrogen bonding) groups
 M number of functional groups
 n number of atoms except hydrogen
 n_p denotes number of data points
 N number of groups, i.e. frequency
 P pressure [kPa]
 S objective function
 T absolute temperature [K]
 τ shearing stress
 η viscosity [P]

Subscripts

ref reference

v reference

b normal boiling point reference

i, j denotes number of a group

k denotes iteration number

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.fluid.2009.02.016](https://doi.org/10.1016/j.fluid.2009.02.016).

Appendix C

C.1. Examples

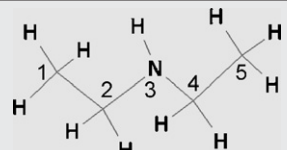
To illustrate the application of the proposed method, a detailed procedure for the estimation of liquid viscosity is given in Tables 28a–28c for:

- N,N-diethylamine.
- Ethylene glycol monopropyl ether.
- Monoethanolamine (MEA).

The latter two examples highlight the application of group interactions. Further examples can be found in the previous publications of this series and can be applied to viscosity with minor modification. In addition, results for T_v and dBv for more than 800 components are given in supplementary material.

Table 28a

Estimation of saturated liquid viscosity curve slope (dBv) and reference temperature (T_v) of N,N-diethylamine and liquid viscosity at 308.15 K.

Component: N,N-diethylamine, number of atoms: 5, $T_{b,exp} = 329.0$ K				
Group	Atoms	Frequency	dBv_i	$T_{v,i}$
1	1,5	2	2×0.0139133	2×89.0803
7	2,4	2	2×0.0213473	2×103.4109
42	3	1	1×0.0136479	1×159.5146
Total sum			0.0841691	544.4970

$$dBv = (0.0841691 / (5^{-2.5635} + 0.0685)) + 3.777 = 4.7713112.$$

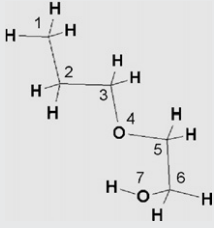
$$T_{v,est} = 21.8444 \times 329.0^{0.5} + (544.4970^{0.9315} / (5^{0.6577} + 4.9259)) - 231.1361 = 210.38 \text{ K}, T_{v,calc} = 207.97 \text{ K} \text{ (From regression to experimental liquid viscosity data stored in the DDB).}$$

$$\eta(dBv, T_{v,calc}) = [e^{(-4.7713112)((308.15-207.97)/(308.15-(207.97/16)))}] \times 13 \text{ mPa s} = 0.2574 \text{ mPa s}.$$

$$\eta(dBv, T_{v,calc}) = [e^{(-4.7713112)((308.15-210.38)/(308.15-(210.38/16)))}] \times 13 \text{ mPa s} = 0.2674 \text{ mPa s}.$$

$$\eta_{exp} = 0.2740 \text{ mPa s} \text{ (A.E. Lutschii, E.M. Obukhova, I.A. Sidorov, Zh. Obshch. Khim., 28 (1958) 2423–2432).}$$

Table 28bEstimation of saturated liquid viscosity curve slope (dBv) and reference temperature (T_v) of ethylene glycol monopropyl ether and liquid viscosity at 318.15 K.

Component: ethylene glycol monopropyl ether, number of atoms: 7, number of interacting groups: 2, $T_{b,exp} = 424.5$ K				
Group	Atoms	Frequency	dBv_i	$T_{v,i}$
1	1	1	1×0.0139133	1×89.0803
4	2	1	1×0.0021727	1×60.3316
7	3,5,6	3	3×0.0213473	3×103.4109
35	7	1	1×0.2490118	1×1078.0840
38	4	1	1×-0.0353055	1×-34.9892
Total sum			0.2938342	1502.7394
Interaction groups				
140	2		2×-0.4239834	2×-643.4378

$$GI_{dBv} = (1/n)((2 \times -0.4239834)/(m-1)) = -0.1211381.$$

$$dBv = ((0.2938342 - 0.1211381)/(7^{-2.5635} + 0.0685)) + 3.777 = 6.0699243.$$

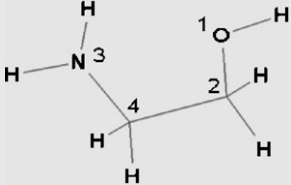
$$GI_{Tv} = (2/n)((2 \times -643.4378)/(m-1)) = -367.679.$$

$$T_{v,est} = 21.8444 \times 424.5^{0.5} + (((1502.7394 - 367.6787)^{0.9315})/(7^{0.6577} + 4.9259)) - 231.1361 = 301.197 \text{ K}.$$

$$\eta(dBv, T_{v,est}) = [e^{(-6.0699243)((318.15-301.197)/(318.15-(301.197/16)))}] \times 1.3 \text{ mPa s} = 0.921814 \text{ mPa s}.$$

$$\eta_{exp} = 1.448 \text{ mPa s (A. Pal, R.K. Bhardwaj, J. Chem. Eng. Data, 46 (4) (2001) 933–938)}.$$

Table 28cEstimation of saturated liquid viscosity curve slope (dBv) and reference temperature (T_v) of monoethanolamine (MEA) and liquid viscosity at 363.15 K.

Component: monoethanolamine, number of atoms: 4, number of interacting groups: 2, $T_{b,exp} = 443.45$ K				
Group	Atoms	Frequency	dBv_i	$T_{v,i}$
7	2,4	2	2×0.0213473	2×103.4109
36	1	1	1×0.2188000	1×1284.7450
40	3	1	1×0.0589131	1×458.7425
Total sum			0.3204077	1950.3093
Interaction groups				
136	2		2×1.031592	2×-41.9608

$$GI_{dBv} = (1/n)((2 \times 1.031592)/(m-1)) = 0.515796.$$

$$dBv = ((0.3204077 - 0.515796)/(4^{-2.5635} + 0.0685)) + 3.777 = 12.3879997.$$

$$GI_{Tv} = (2/n)((2 \times -41.9608)/(m-1)) = -41.9608.$$

$$T_{v,est} = 21.8444 \times 443.45^{0.5} + (((1950.3093 - 41.9608)^{0.9315})/(4^{0.6577} + 4.9259)) - 231.1361 = 382.276 \text{ K}.$$

$$\eta(dBv, T_{v,est}) = [e^{(-12.3879997)((363.15-382.276)/(363.15-(382.276/16)))}] \times 1.3 \text{ mPa s} = 2.6137 \text{ mPa s}.$$

$$\eta_{exp} = 2.337 \text{ mPa s (J. Aguila-Hernandez, A. Trejo, B.E. Garcia-Flores, R. Molnar, Fluid Phase Equilib., 267 (2) (2008) 172–180)}.$$

C.2. Derivation of Eq. (6)

Replacing the parameter C in Eq. (3) by the generalized expression (Eq. (5)) yields:

$$\ln\left(\frac{\eta}{\eta_{ref}}\right) = A\left(\frac{B}{T - (T_v/s)}\right)$$

The difference between $\ln(\eta/\eta_{ref})$ at T and $\ln(1.3cP/\eta_{ref})$ at T_v is therefore:

$$\ln\frac{\eta}{\eta_{ref}} - \ln\frac{1.3cP}{\eta_{ref}} = \ln\frac{\eta}{1.3cP} = A + \frac{B}{T - (T_v/s)} - A - \frac{B}{T_v - (T_v/s)}$$

Simplification leads to:

$$\begin{aligned} \ln\frac{\eta}{1.3cP} &= B\left[\frac{1}{T - (T_v/s)} - \frac{1}{T_v - (T_v/s)}\right] \\ &= B\left[\frac{T_v - T}{(T - (T_v/s))(T_v - (T_v/s))}\right] \end{aligned}$$

With $dBv = B/(T_v - T_v/s)$ and $s = 16$, this leads to Eq. (6):

$$\ln\frac{\eta}{1.3cP} = -dBv\left[\frac{T - T_v}{(T - (T_v/16))}\right]$$

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